

Useful factors are fewer than you think ^{*}

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Abstract

We examine how many factors out of a wide range of 207 have incremental information in explaining cross-sectional stock returns. First, we find that the significance of each factor changes drastically over time. After accounting for the false discovery rate (FDR), only 157 out of 207 factors are significant from 1967 to 2021, and only 56 from 2000 to 2021. Second, from 2000 to 2021, we find strikingly that only 3 clusters of factors have incremental information. We further propose a new flexible time-varying latent factor model, and test in an alternative way on the number of factors that capture the information of the 56 significant factors while controlling for FDR, and find only 3, the market plus 2 latent ones, when without asset pricing restrictions, and find only 4, the market plus 3 latent ones, when with asset pricing restrictions, respectively. In either case, the number of factors is much fewer than widely believed.

JEL Classifications: C12, C55, G12

Key words: Cross-sectional returns, Factor models, False discovery rate, Multiple testing, Time-varying.

1 Introduction

Numerous factors have been identified to explain the cross-section of stock returns in the past 50 years. In his presidential address, [Cochrane \(2011\)](#) asks two important and related questions. First, how many factors do we really need? Second, given a factor model such as the CAPM, are there other factors that can provide incremental information for explaining the cross-section of expected stock returns?

Answers to the questions are inconclusive. On the one hand, [Harvey et al. \(2016\)](#) argue that most claimed findings in significant factors for the cross-sectional stock returns are likely false after controlling for the false discovery rate (FDR), and [McLean and Pontiff \(2016\)](#) show that the out-of-sample and post-publication return are substantially lower. Both studies allow for a large number of factors, although their studies cast doubt on

the validity of many claimed factors. On the other hand, [Chen \(2021\)](#) shows that p -hacking cannot explain the large set of observed factors, and [Chen and Zimmermann \(2021\)](#) replicate up to 207 factors, the largest to date, and will be ones analyzed further by this paper. Consistent with [Chen and Zimmermann \(2021\)](#), [Jensen et al. \(2023\)](#) show that most anomalies originally discovered in the US are robust internationally.

In this paper, we address two questions. First, inspired by [Harvey et al. \(2016\)](#), we ask how many of the 207 factors remain significant after controlling for FDR. Complimenting to [Harvey et al. \(2016\)](#), we examine not only far more factors but also use additional statistical procedures and consider weighting factors by their economic motivations. Moreover, we study how the significance changes over time, and the number of economic risk sources as captured by factor clusters.

The second question addressed here is how many economic risk sources we need to capture the risks of a given set of significant factors. For this purpose, we propose a flexible high-dimensional time-varying latent factor model to test the number of factors. While latent factor models are used in finance theoretically since [Ross \(1976\)](#), and empirically by [Connor and Korajczyk \(1986\)](#) and [Lettau and Pelger \(2020b\)](#), among many others, we find that the constant loading models are rejected by the data. Hence, our proposed time-varying latent factor model also adds to the literature on factor models.

Empirically, we find that, of the comprehensive set of 207 factors compiled by [Chen and Zimmermann \(2021\)](#), there are only 157 significant ones after accounting for FDR, from 1967 to 2021, and there are only 56 from 2000 to 2021. A 20-year rolling estimation shows a declining pattern of the number of significant factors. Not all factors are economically well established. Of the 15 factors except for the market that are motivated by well-known factor models of Fama and French (1993, 2015, 2018), [Hou et al. \(2015\)](#), [Stambaugh and Yuan \(2017\)](#), and [Daniel et al. \(2020\)](#), we find that, while 14 of them are significant from 1967 to 2021, there are only 3 of them remain significant from 2000 to 2021. The decline of such model-motivated (MM) factors is simply striking. Moreover, for the 56 significant factors from 2000 to 2021, they are highly correlated and many of them are redundant. In terms of clusters, 3 of them are sufficient to capture the variations of the 56 factors.

Consistent with the above findings, our flexible time-varying latent factor model also reveals, in the case of without imposing asset pricing restrictions in the latent factor model, that there are three factors, the market plus 2 latent factors, which are sufficient to explain all 56 significant factors. To gain improved explanatory power of the latent factor model, we also consider the case of imposing asset pricing restrictions in a manner similar to Lettau and Pelger (2020a, b), who impose such constraints on the PCA. In this case, we find that there are a total of 4 factors that are needed, the market plus 3 latent factors. Overall, our conclusion is that the number of systematic risk sources is much fewer than commonly believed.

Our paper shows a sharp decline in factors from 1967 to 2000. Why is the number of factors declining over time? There appear three major reasons. First, as put forth by [Green et al. \(2017\)](#), there are some major regulatory changes around or after the year 2000: the Sarbanes-Oxley Act, the acceleration of 10-Q and 10-K filing requirements by the SEC, and the introduction of autoquoting by the NYSE. All of these lead to falling costs and faster speed for exploiting mispricing. Second, [Schwert \(2003\)](#) and [McLean and Pontiff \(2016\)](#), among others, argue forcefully that anomalies, such as the size and value, are often attenuated or disappear after the original research was published, due to learning by arbitragers and investors. Third, [Lo \(2004\)](#) proposes his adaptive market hypothesis that investors and the market adapt to a changing environment over time. [Nagel \(2021\)](#) further emphasize that adaptivity is a feature in the age of big financial data. Consistent with the adaptive market hypothesis, many mispricing factors that impact firms' profitability are likely understood and exploited away over time by investors, and hence are no longer as important as they used to be. Some features are subject to structural change around 2000, our analysis below will focus on the 2000 to 2021 period.

Our paper is related to a growing number of recent studies on factors and firm characteristics. [Kelly et al. \(2019\)](#) introduce instrumented PCA (IPCA) to model latent factors and time-varying loadings. In contrast to their model, our loadings are nonparametric and we focus on testing the number of factors in replacing the existing factor candidates. While [Freyberger et al. \(2020\)](#), [Gu et al. \(2020\)](#), [Kozak et al. \(2020\)](#) and [Avramov et al. \(2022\)](#), among others, also provide insights on how many factors or firm characteristics

that matter, they focus on predicting the expected returns using past information. In contrast, we focus on factor models explaining the expected returns with contemporaneous information. The difference between the two is obvious: while the market factor can explain the expected returns well in many applications, it is of little use as a predictor. Although [He et al. \(2022\)](#) and [Chib et al. \(2022\)](#) achieve factor reduction in the explanatory setting, they apply neither clusters nor latent factors. As a result, they reach different conclusions from us.

The rest of the paper is organized as follows. In Section 2, we consider several models and testing procedures including the multiple testing procedure based on the time-varying factor model, which has never been considered in the literature. Section 3 discusses the empirical findings using the long-short strategy returns. In Section 4, we compare the accumulative wealth of investment based on the time-varying latent factor model, the cluster factors and the benchmark Fama-French 3-factor model. Section 5 concludes. All mathematical proofs are provided in the Online Appendix.

2 Methodology

2.1 Significant factors under FDR

Suppose we have N factors formed by long-short portfolios (see, e.g., [Chen and Zimmermann \(2021\)](#)). Consider the CAPM regressions,

$$r_{i,t} = \alpha_i + \beta_i f_t + u_{i,t}, \quad 1 \leq i \leq N, 1 \leq t \leq T, \quad (1)$$

where r_{it} is the excess return of long-short portfolios i at time t , f_t is the market excess return, β_i is the factor loading, and $u_{i,t}$ is the idiosyncratic component.

The objective is to find those factors with truly significant alphas after controlling for FDR. To do so, we formulate the following hypotheses:

$$\mathbf{H}_0^i : \alpha_i = 0 \quad \text{versus} \quad \mathbf{H}_A^i : \alpha_i \neq 0, \quad \text{for} \quad 1 \leq i \leq N. \quad (2)$$

This is a multiple-testing problem. The central difficulty when testing a large number of null hypotheses is that several of them may appear to be rejected, purely by chance. We would like the set of rejected null hypotheses to have high precision so that most discovered risk factors are truly useful for prediction. It is known that separately controlling the FDR for each individual test does not provide any guarantee of precision. Thus, we adopt the popular [Benjamini and Hochberg \(1995\)](#) (BH) method to control the FDR at level q as follows:

1. Run time series regressions and obtain the OLS estimator $\widehat{\beta}_i$.
2. Estimate α_i by subtracting the estimated risk premium from average returns:

$$\widehat{\alpha}_i = \bar{r}_i - \widehat{\beta}_i^\top \bar{f}.$$

3. For each $1 \leq i \leq N$, compute the individual t-statistics t_i for the hypothesis $\mathbb{H}_0^i$ and the corresponding p-values p_i . Sort the p-values and denote $p_{(1)} \leq p_{(2)} \leq \dots \leq p_{(N)}$ as the ordered p-values.
4. Choose

$$\widehat{k} = \max \left\{ i : p_{(i)} \leq \frac{iq}{N} \right\},$$

and reject all null hypotheses $\mathbb{H}_0^i$ for $i = 1, \dots, \widehat{k}$.

2.2 Incorporating economic importance

[Chib et al. \(2022\)](#) show that a seven-factor asset pricing model, consisting of {Mkt, SMB, MOM, ROE, MGMT, PERF, and PEAD} gets the most support from the data. We divide the set of hypotheses into two groups. One consists of {Mom12m, Mom12mOffSeason, Mom6m, MomOffSeason, MomOffSeason06YrPlus, MomOffSeason16YrPlus, MomRev, MomSeason, MomSeason06YrPlus, MomSeason11YrPlus, MomSeason16YrPlus, MomSeasonShort, MomVol, Investment, RoE}, which are factors that cover the markets and fundamentals and correspond to the factors with stronger data support. {Mom12m, Mom12mOffSeason, Mom6m, MomOffSeason, MomOffSeason06YrPlus, MomOffSeason16YrPlus, MomRev,

MomSeason, MomSeason06YrPlus, MomSeason11YrPlus, MomSeason16YrPlus, MomSeasonShort, MomVol} are price momentum factors. RoE is the return on equity factor, and Investment is a fundamental factor that measures the investment of a firm. The other group consists of the rest of the factors.

The group of model-motivated factors is expected to be rejected more easily. To understand the difference between the two groups, we adopt the procedure proposed by [Genovese et al. \(2006\)](#) (GRW) to assign weights to individual hypotheses and conduct multiple hypothesis testing while incorporating prior information about the hypotheses:

1. Assign weights so that $\sum_{i=1}^N w_i = N$.
2. For each $i = 1, \dots, N$, compute $p_i^w = p_i/w_i$.
3. Apply [Benjamini and Hochberg \(1995\)](#) at level q to p_i^w .

We consider a binary weighting scheme. Define a confidence parameter γ , which measures how confident we are in the group with model-motivated factors vs the other group. For example, if $\gamma = 1$, we are indifferent between these two groups; if $\gamma = 2$, we believe that the important group is twice more likely to be identified than the other group. Let G_I denote the group with economically important factors, G_R denote the group with the remaining factors, and N_I be the number of factors in G_I . The binary weighting is chosen as:

$$w_i = \begin{cases} \frac{N\gamma}{N+(\gamma-1)N_I} & \text{if } i \in G_I, \\ \frac{N}{N+(\gamma-1)N_I} & \text{otherwise.} \end{cases} \quad (3)$$

Note that the normalization restriction $\sum_{i=1}^N w_i = N$ is satisfied by construction.

2.3 Are our results robust to latent factors?

Model (1) assumes that f_t is the market excess return, which is observable. Our empirical analysis shows that the anomaly returns based on model (1) have high cross-sectional correlation¹, suggesting that anomalies potentially have exposures to unknown common risk factors. As discussed in [Giglio and Xiu \(2021\)](#) and [Giglio et al. \(2021\)](#), uncounted

¹Around half of the correlations among the estimated anomaly returns exceed 0.5.

latent factors might lead to biased alpha estimates and misleading testing results. Therefore, we reconsider model (1), assuming f_t is a r -dimensional vector of factors, which is a collection of the observed factor $f_{o,t}$ and latent factors $f_{l,t}$. Model (1) can be rewritten as

$$r_{i,t} = \underbrace{\alpha_i + \beta_i^\top \lambda}_{\text{Return}} + \underbrace{\beta_i^\top (f_t - \mathbb{E}[f_t])}_{\text{Risk}} + u_{i,t}, \quad i \leq N, t \leq T, \quad (4)$$

where the loading β_i is interpreted as exposure to systematic risk factors, and λ as the risk premiums associated with factors. Note that $\alpha_i, \beta_i, \lambda$ and $f_{l,t}$ are all unobserved and need to be estimated from data. We adopt [Giglio et al. \(2021\)](#)'s procedure to compute the bootstrap p-value p_i and then apply [Benjamini and Hochberg \(1995\)](#).

2.4 Time-varying alphas

The rolling regression in Section 2.1 shows that both α and β appear to change over time. We further apply [Fu et al. \(2023\)](#) to test whether α and β are constant over time and the p -value based on 1,000 bootstrap iterations is 0.003, which suggests strong rejection at any conventional significance levels. Therefore, we consider a time-varying high-dimensional factor model for excess returns of the form

$$r_{i,t} = \alpha_{i,t} + \beta_{i,t}^\top \lambda_t + \beta_{i,t}^\top (f_t - \mathbb{E}[f_t]) + u_{i,t}, \quad i \leq N, t \leq T, \quad (5)$$

where $f_t = (f_{o,t}^\top, f_{l,t}^\top)^\top$ is again a collection of the observed tradable factors $f_{o,t}$ and latent factors $f_{l,t}$. But unlike model (4), $\alpha_{i,t}$, $\beta_{i,t}$ and λ_t are allowed to be changing over time. The number of factors is time-invariant. This model can be viewed as a generalization of the constant parameter factor model considered in [Giglio et al. \(2021\)](#) by allowing for structural changes in alpha and factor loadings.

To cover a wide range of potential time variation, we follow the literature on smooth time-varying parameter models (e.g., [Robinson \(1989\)](#), [Cai \(2007\)](#), [Su and Wang \(2017\)](#))

and model $\alpha_{i,t}$ and $\beta_{i,t}$ as non-stochastic functions of t/T , that is,

$$\begin{aligned}\alpha_{i,t} &= \alpha_i(t/T), \\ \beta_{i,t} &= \beta_i(t/T),\end{aligned}$$

where $\alpha_{i,t}(\cdot)$ and $\beta_{i,t}(\cdot)$ are some unknown smooth functions of t/T on $[0, 1]$ for each i . The specification that alpha and beta are some functions of ratio t/T rather than time t only is a common scaling scheme in the literature (see, e.g., [Phillips and Hansen \(1990\)](#), [Robinson \(1991\)](#) and [Cai \(2007\)](#)). The reason for this specification is that nonparametric estimators for $\alpha_{i,t}(\cdot)$ and $\beta_{i,t}(\cdot)$ will not be consistent unless the amount of data on which they depend increases, and merely increasing the sample size will not necessarily improve the estimation of $\alpha_{i,t}(\cdot)$ and $\beta_{i,t}(\cdot)$ at some fixed point t , even if some smoothness conditions are imposed. The amount of local information must increase suitably if the variance and bias of nonparametric estimators of α_t and β_t are to decrease suitably.

To estimate the model (5), we first consider a local least squared estimation and obtain the time-varying estimator $\widehat{\beta}_{o,i,t}$ for the loading of the observed factor and residual $z_{i,t}$

$$\widehat{\beta}_{o,i,t} = \left(\sum_{s=1}^T K_{h,ts} (f_{o,s} - \bar{f}_{o,t})(f_{o,s} - \bar{f}_{o,t})^\top \right)^{-1} \left(\sum_{s=1}^T K_{h,ts} (r_{i,s} - \bar{r}_{i,t})(f_{o,s} - \bar{f}_{o,t}) \right),$$

and

$$z_{i,t} = (r_{i,t} - \bar{r}_{i,t}) - \widehat{\beta}_{o,i,t}^\top (f_{o,t} - \bar{f}_{o,t}),$$

where

$$\bar{f}_{o,t} = \frac{1}{T} \sum_{s=1}^T K_{h,ts} f_{o,s}$$

and

$$\bar{r}_{i,t} = \frac{1}{T} \sum_{s=1}^T K_{h,ts} r_{i,s}$$

are the local sample averages of the observed factor and return at time t respectively. $K_{h,ts}$

is the boundary-modified kernel function:

$$K_{h,ts} = \begin{cases} h^{-1}k(\frac{t-s}{Th})/\int_{-s/(Th)}^1 k(u) du & \text{if } s \in [1, \lfloor Th \rfloor], \\ h^{-1}k(\frac{t-s}{Th}) & \text{if } s \in [\lfloor Th \rfloor, T - \lfloor Th \rfloor], \\ h^{-1}k(\frac{t-s}{Th})/\int_{-1}^{(1-s/T)/h} k(u) du & \text{otherwise,} \end{cases} \quad (6)$$

the kernel $k(\cdot) : [-1, 1] \mapsto \mathbb{R}^+$ is a prespecified symmetric probability density, $h = h(T, N)$ is a bandwidth parameter, and $\lfloor Th \rfloor$ denotes the integer part of Th . Examples of $k(\cdot)$ include the uniform, Epanechnikov and quartic kernels. This boundary-modified kernel function has been used in [Hong and Li \(2005\)](#) and [Su and Wang \(2017\)](#).

Then we define the local sample covariance matrix as

$$\widehat{\Sigma}_t = \frac{1}{T} \sum_{s=1}^T K_{h,ts} \mathbf{z}_s \mathbf{z}_s^\top \quad (7)$$

where $\mathbf{z}_s = (z_{1,s}, z_{2,s}, \dots, z_{N,s})^\top$. Under the identification condition $\frac{1}{N} \boldsymbol{\beta}_{l,t}^\top \boldsymbol{\beta}_{l,t} = I$, the L-RP-PCA estimator $\widehat{\boldsymbol{\beta}}_{l,t}$ for the loading of the latent factor is comprised of $\sqrt{N}$ times top r eigenvectors of $\widehat{\Sigma}_t$ in descending order by corresponding eigenvalues.

Similar to [Fan et al. \(2022\)](#) and [Giglio et al. \(2021\)](#), we run a cross-sectional regression of $\bar{r}_t$ on $\widehat{\boldsymbol{\beta}}_t$ and a constant regressor 1_N to obtain the risk premia of latent factors:

$$\widehat{\boldsymbol{\lambda}}_{l,t} = \left(\widehat{\boldsymbol{\beta}}_{l,t}^\top \mathbb{M}_{1_N} \widehat{\boldsymbol{\beta}}_{l,t} \right)^{-1} \left(\widehat{\boldsymbol{\beta}}_{l,t}^\top \mathbb{M}_{1_N} \bar{r}_{\lambda,t} \right), \quad (8)$$

where

- $\bar{r}_{\lambda,t} = \bar{r}_t - \widehat{\boldsymbol{\beta}}_{o,t}^\top \bar{f}_{o,t}$;
- $\mathbb{M}_{1_N} = I_N - 1_N (1_N^\top 1_N)^{-1} 1_N^\top$, and 1_N is the $N \times 1$ vector of ones, I_p is an identity matrix.

And the mispricing $\widehat{\alpha}_t$ is obtained via

$$\widehat{\alpha}_t = \bar{r}_t - \widehat{\boldsymbol{\beta}}_t^\top \widehat{\boldsymbol{\lambda}}_t, \quad (9)$$

where $\widehat{\boldsymbol{\beta}}_t = (\widehat{\boldsymbol{\beta}}_{o,t}^\top, \widehat{\boldsymbol{\beta}}_{l,t}^\top)^\top$ and $\widehat{\boldsymbol{\lambda}}_t = (\bar{f}_{o,t}^\top, \widehat{\boldsymbol{\lambda}}_{l,t}^\top)^\top$. As shown in Theorem 2 in the appendix, $\widehat{\alpha}_t$

has an asymptotic normal distribution and hence the t-statistic and p -values based on the limiting distribution can be calculated. To improve the finite sample performance, we consider the following bootstrap procedure in the same spirit as [Giglio et al. \(2021\)](#).

1. The above two-step nonparametric estimation yields the residual

$$\widehat{u}_{i,t} = r_{i,t} - \bar{r}_{i,t} - \widehat{\beta}_{i,t}^\top \widehat{v}_t,$$

where $\widehat{v}_t = \left((f_{o,t} - \bar{f}_{o,t})^\top, \widehat{v}_{l,t}^\top \right)^\top$, and $\widehat{v}_{l,t} = \frac{1}{N} \widehat{\beta}_{l,t}^\top \mathbf{z}_t$. Then obtain a wild bootstrap residual $\widehat{u}_{i,t}^* = a_{i,t} \widehat{u}_{i,t}$, where $\{a_{i,t}\}$ is a sequence of i.i.d. random variables with mean 0 variance 1, and $\widehat{\bar{u}}_{i,t} = \widehat{u}_{i,t} - T^{-1} \sum_{t=1}^T \widehat{u}_{i,t}$. And construct a bootstrap sample

$$r_{i,t}^* = \widehat{\beta}_{i,t}^\top \widehat{\lambda}_t + \widehat{\beta}_{i,t}^\top \widehat{v}_t + \widehat{u}_{i,t}^*.$$

2. Estimate the $\widehat{\beta}_{i,t}^*$ via local LS:

$$\widehat{\beta}_{i,t}^* = \left(\sum_{s=1}^T K_{h,ts} (\mathbf{v}_s - \bar{\mathbf{v}}_t) (\mathbf{v}_s - \bar{\mathbf{v}}_t)^\top \right)^{-1} \left(\sum_{s=1}^T K_{h,ts} (r_{i,s}^* - \bar{r}_{i,t}^*) (\mathbf{v}_s - \bar{\mathbf{v}}_t) \right),$$

3. Obtain $\widehat{\lambda}_t^*$ and the risk premium $\widehat{\alpha}_t^*$ for the bootstrap sample

$$\widehat{\lambda}_t^* = \left(\widehat{\beta}_t^{*\top} \mathbb{M}_{1_N} \widehat{\beta}_t^* \right)^{-1} \left(\widehat{\beta}_t^{*\top} \mathbb{M}_{1_N} \bar{r}_t^* \right),$$

and

$$\widehat{\alpha}_t^* = \bar{r}_t^* - \widehat{\beta}_t^{*\top} \widehat{\lambda}_t^*.$$

4. Repeat step 1-3 for B times and compute the bootstrap p-values

$$p_{i,t} = \frac{1}{B} \sum_{b=1}^B 1 \left(\widehat{\alpha}_{i,t,b}^* > \widehat{\alpha}_{i,t} \right).$$

The above procedure assumes that the dimension r_l of the latent factor $f_{l,t}$ is known. However, in practice, we need to estimate r_l as well. We extend the eigenvalue ratio-

based estimator, proposed by [Ahn and Horenstein \(2013\)](#). Let $\widehat{\phi}_{1,t} \geq \widehat{\phi}_{2,t} \geq \dots \geq \widehat{\phi}_{r,t} \geq 0$ be the ordered eigenvalues of $\widehat{\Sigma}_t$. And the generalized eigenvalue ratio-based estimator for r can be defined as

$$\widehat{r}_l = \operatorname{argmax}_{1 \leq j \leq r_{\max}} \frac{1}{T} \sum_{t=1}^T \frac{\widehat{\phi}_{j,t}}{\widehat{\phi}_{j+1,t}}, \quad (10)$$

and $r_{\max}$ is a selected upper bound. We could choose $r_{\max} = \lfloor N/2 \rfloor$ or $r_{\max} = \lfloor N/3 \rfloor$, following [Ahn and Horenstein \(2013\)](#). The consistency of $\widehat{r}_l$ is confirmed in Theorem 3 in the appendix. Other estimators such as a BIC-based estimator can be applied as well.

3 Empirical Results

3.1 Data Source

Data for this paper is obtained from two sources: open source asset pricing and the Fama-French data library. Open source asset pricing database is constructed by [Chen and Zimmermann \(2021\)](#), from which we collect the monthly data of long-short strategy returns of 207 factors sampled from 1967 to 2021. And monthly market excess return is obtained from the Fama-French data library.

3.2 Significant factors in CAPM under FDR control

We follow the CAPM regression and BH procedure specified in Section 2.1 on two samples differentiated by sampling period: the first one is a full sample from 1967 to 2021, and the other sample is more recent with a sampling period from 2000 to 2021. In the full sample, the observations of some factors start from some years later than 1967. In the recent sample, the observations of some factors end before 2021. These observations are kept as we will conduct a rolling estimation that evaluates the sample size effect in this section.

Table 1 reports the significant factors in the full sample. After the BH procedure is implemented, 157 out of 207 factors get rejected. Among them, *DivSeason* is the most significant factor with t-stat over 16.

The results on the second sample are presented in Table 2. The estimation with more recent data induces a plunge in the number of rejections. In total, under the same FDR control, only 56 out of 207 factors are rejected, yielding a decline of nearly 100 significant factors compared with the first sample. In addition, AnnouncementReturn takes the place of DivSeason and becomes the most significant factor with t stat 9.44, which is much less than the t-stat of DivSeason on the full sample. Results on the full sample and recent sample suggest that significant factors experience a striking loss.

One possibility of this phenomenon stems from the sample size argument: in the full sample, the observations for each factor are around 2.5 times more than the ones in the recent sample. A rolling-window estimation is conducted to investigate the potential sample size effect. In the rolling-window study, we construct samples starting from 1967/01 with a fixed sample size of 20 years, rolling by month. In total, there are 421 samples constructed with samples starting years from 1967 to 2002. Factors are deleted if there are missing observations in order to control the sample size. The BH FDR control method is applied to each sample, allowing to capture of the trend of the number of rejected factors across time. Figure 1 present the number of rejections across time. The number of rejections follows a hump-like shape: it increases at the beginning and then begins to decrease with the turning point around 1981. The results from the rolling-window study dispute the argument that the decreasing sample size induces fewer numbers of rejections in the recent sample.

3.3 Model-motivated factors

We follow the GRW procedure specified in Section 2.2 on model-motivated and on full sample and recent sample, respectively. By the nature of the GRW procedure, weights higher than 1 will boost the corresponding p-values so that they could be more likely to be rejected via the BH multiple-test procedure. Since the p-value of *investment* is too large in the full sample, it cannot be rejected for any value of γ . Therefore, the procedure is applied only to the recent sample. Table 4 presents the rejection results with $\gamma = 2, 5$, and 10 in the recent sample. The number of identifications in G_I increases from 3 to 5 and then to 8 as γ increases from 2 to 5 and then 10.

3.4 Factors in latent factor models

The procedure specified in Section 2.3 with 3 latent factors as well as the market factor and with bootstrap size equal to 1,000 is conducted. The results are reported in Table 5 and Table 6. The similarity of the results between the OLS model and the latent factor model is observed. In the full sample, 150 out of 207 factors are identified under the BH test, which is close to the result obtained via CAPM (157). In the recent sample, the number of rejected factors drops from 150 to 44, while in CAPM this number is 56. Therefore, the factor model shows an even sharper decrease in the number of identified factors. Among the model-motivated factors, in the full sample, only *Investment* fails to be identified, while only 3 important factors are identified in the recent sample. We note that the important factors identified by the latent factor model are in line with the results obtained via CAPM in Section 2.1.

A rolling study with a 20-year rolling window is also conducted. The results are shown in Figure 3. We observe that the number of rejections rises in fluctuation in the period from 1967 to around 1985. Then it begins to decrease and falls to the bottom at around 40 in around 2000. Combining all these findings from the latent factor model strengthens the robustness of the results obtained in Section 3.2.

3.5 Factors in time-varying factor model

This section discusses the results obtained from the time-varying factor model specified in Section 2.4. To begin with, we applied the generalized eigenvalue-ratio procedure in (10) to the complete dataset spanning from 1967 to 2021. Our analysis reveals that two latent factors effectively account for the majority of factors after controlling for the impact of the market factor in the dataset. Accordingly, we estimate the time-varying model in (5) using two latent factors and one market factor. We generate heat maps of the alphas estimated using the time-varying factor model for all factors and model-motivated factors, which are shown in Figure 4 and Figure 5, respectively. These heat maps demonstrate that the color density changes smoothly over time, indicating the time-varying nature of the evolution of the alphas. This provides justification for the application of the

time-varying latent factor model to our dataset.

The wild bootstrap procedure, as outlined in Section 2.4, is implemented with bootstrap size $B = 1,000$ to obtain the p-value of the alpha estimate of each factor at each time point. And the BH procedure with FDR control at 1 percent is conducted and the corresponding results are presented in Figure 6. The shape of the rejection curve is similar to Figure 1 and Figure 3 that we obtain via the rolling study using the CAPM model and the constant parameter latent factor model, respectively, albeit with some variations in the level of rejection counts. Note that we use the kernel estimation in the time-varying model that obtains estimations using a local, two-sided sample, while in the rolling study in CAPM and constant parameter latent factor model, the rolling sample is one-sided and taken from the date labeled on the axis. Given that, the timing of the peak of the factor identifications of these three models also coincides. We also note that the number of identified factors starts at approximately 90, and the number increases to the peak around 1993, followed by a sharp decline to around 20 by the end of the period. Figure 7 shows the rejections curve of economically important factors identified using the time-varying factor model, which exhibits a similar pattern to that for all factors shown in Figure 6. The peak number of identified factors reaches 13 out of 15 and drops to nearly 0 by the end of the study period, reinforcing our previous findings that there are fewer useful factors than we thought and the number of useful factors is declining over time.

We compare the explanatory power of the identified latent factors with that of the Fama-French 3 factors (ff3) and Fama-French 5 factors (ff5) by applying the time-varying factor model (5) with either ff3 or ff5 or the identified latent factors. Our results show that for the long-short portfolio, the market factor plus two latent factors provide the most significant explanation for the variation in returns, with an R^2 that is 69% higher than that of ff3 and 46% higher than that of ff5.

We further conduct a cluster analysis on the factors identified from CAPM applied to the full sample and to the recent sample. We use the correlation between factors as the distance measure and evaluate aggregation levels across different numbers of clusters, as shown in Figure 8 and in Figure 9 for full sample and recent sample, respectively. The y-axis of the plot, named Height, is a measure of intergroup dissimilarity. We find

that among the number of clusters ranging from 2 to 9, the clustering partitions implement the most effective reduction in intergroup dissimilarity when 5 clusters are selected, while the analogical number for the recent sample is 3. This result coincides with the number of factors estimated through the generalized eigenvalue-ratio procedure, which reinforces the robustness of the selection of latent factor numbers. The assignment of factors to each cluster is shown in Table 8, while the assignment of model-motivated factors is displayed in Table 10. The correlation matrix among identified factors, sorted by cluster, is visualized in Figure 10. As observed from the figure, factors in each cluster exhibit strong correlation, while factors in different clusters are weakly correlated. Despite the large number of identified factors in the whole sample period, many factors are correlated with each other, and only three factors (the market factor and two latent factors) would be sufficient to explain the major variation.

4 Dicussion

4.1 Out-of-sample and Post-publication return study

As argued by McLean and Pontiff (2016), the out-of-sample and post-publication returns are substantially and significantly lower. Inspired by their work, we conduct a study of out-of-sample and post-publication returns of each factor. First, CAPM is run for the long-short strategy return of each factor before and after publication. The t-stats of the intercept are compared. For each factor, the regression is run twice with the sample before publication and after publication. In order to alleviate the issue induced by imbalanced sample size, the samples before publication or after publication are truncated so that they have the same size. The t -stats of α_i are collected and compared, which are presented in Table 11. Among 207 factors, 157 factors experienced significant decay. *dNoa* has its significance level affected the most: the t-stat dropped by 7.15 after the publication. In addition, among these factors, 72 factors have their significance level drop from above 1.96 to below 1.96, which is the threshold for the 5-percent-size individual t-test. Also, we conduct BH test procedures on all factors before and after publication, and the results

are displayed in Table 12. Not surprisingly, the number of identified factors witnesses a sharp decline after publication: before publication, 112 out of 207 factors are being identified, while only 30 factors can be identified by BH procedure after publication.

Following McLean and Pontiff (2016), we explore the statistical robustness of post-publication decay. We run the following panel regression:

$$R_{i,t} = \alpha_i + \beta_1 \text{Out-of-sample}_{i,t} + \beta_2 \text{Post-publication}_{i,t} + \varepsilon_{i,t},$$

where $\text{Out-of-sample}_{i,t}$ is a dummy variable equal to 1 when the sample at time t is after the sample period in the original paper of factor i , and $\text{Post-publication}_{i,t}$ is a dummy variable equal to 1 when the sample at time t is after the publication date of factor i . The results are shown in Table 13, which shows a strong negative correlation between out-of-sample and the return, and between post-publication and the return. On average, the long-short strategy of each factor declines by around 0.3 percentage after the publication and after the sampling period in the original paper.

4.2 Extracting Factors using LRP-PCA

In this section, we explore factors that can capture the information of not only in the covariance of anomalies, but also are capable of explaining their expected returns for the recent data from 2000 to 2021. To achieve this, we consider a time-varying factor model without an intercept term:

$$r_{i,t} = \beta_{i,t}^\top \mathbf{f}_t + u_{i,t}, \quad (11)$$

where $\mathbf{f}_t \in \mathbb{R}^r$ represents latent factors, and $\beta_{i,t}$ the corresponding factor loading indexed by t . To enable $\mathbf{f}_t$ to capture both the covariance and the expected returns of anomalies, we consider a local risk-premium PCA (LRP-PCA), which extends the RP-PCA (Lettau and Pelger (2020a), Lettau and Pelger (2020b)) to the time-varying latent factor models. The estimation involves the following steps:

1. Calculate the statistic that aggregates the local information in both first and second

moments:

$$\widehat{\Sigma}_{t,\gamma} = \frac{1}{T} \sum_{s=1}^T K_{h,ts} (r_s r_s^\top + \gamma \bar{r}_s \bar{r}_s^\top),$$

where $\bar{r}_s$ equals the local mean of r_t at time s defined in Equation 6 and $K_{h,ts}$ is some boundary-adjusted kernel defined in Equation 6, and $\gamma \geq -1$ is a hypoparameter that balances the information of the first and second moments.

2. The LRP-PCA estimator $\widehat{\beta}_t$ is comprised of $\sqrt{N}$ times top r eigenvectors of $\widehat{\Sigma}_{t,\gamma}$ in descending order by eigenvalues.
3. Obtain the least square estimate of factors f_t :

$$\widehat{f}_t = \frac{1}{N} \widehat{\beta}_t^\top r_t.$$

The number of latent factors is set to 3 in alignment with the number of clusters identified in section 3.5. γ is set to 1 to better capture the expected returns of anomalies.

We assess the performance of the factor set composed of 3 LRP-PCA latent factors and the market factor, and compare it with the Fama-French factors in explaining the anomalies. Analogous to Section 2.1, a series of regressions of anomalies on 3 latent factors plus market factor are conducted to examine the significance of these anomalies after controlling for FDR at one percent. The results of 3 latent factors, FF3 and FF5 are shown in Table 15. Notably, LRP-PCA latent factors plus the market factor exhibit superior explanatory power for anomalies compared to both FF3 and FF5 factors, characterized by fewest significant anomalies, lowest root mean square of alphas, and the highest R^2 .

Next, we evaluate the extent to which Fama-French 5 factors can be explained by 3 latent factors and the market factor, and vice versa. Regressions of FF5 factors on latent factors and the market factor, as well as the reverse regressions, are conducted, with the results presented in Table 16. After controlling for FDR, none of the FF5 factors are significant when regressed on latent factors and the market factor, meaning that all of them can be explained by the LRP-PCA factors and the market factor.

Conversely, there are only one latent factor can be explained well by the FF3 factors. Therefore, latent factors from LRP-PCA plus the market factor not only explain anomalies

but also provide a comprehensive explanation for the FF5 factors, whereas Fama-French factors exhibit less effectiveness in this regard.

Finally, we compare the Sharpe ratio of the optimal portfolios constructed by 3 latent factors and market factor, FF3 factors and FF5 factors, respectively. As argued by [Barillas and Shanken \(2017\)](#), comparing two factor models is to compare their Sharpe ratios of the optimal portfolio formed out of each set of the factors. Given a set of factors, the optimal portfolio is obtained by solving the following optimization problems:

$$w_{op} = \arg \max_w w^\top \mu - \frac{1}{3} w^\top \Sigma w,$$

where μ is the vector of expected returns of factors and Σ is the covariance matrix of factors. The findings shown in Table 17 suggest that among the three sets of factors, portfolios constructed with 3 latent factors and the market factor exhibit the highest Sharpe ratio.

5 Conclusion

Our paper investigates the number of factors that remain statistically significant under FDR control since 1967 and since 2000, respectively. We find a sharp decline in their numbers. We verify this finding using the comprehensive factor data set constructed by [Chen and Zimmermann \(2021\)](#). Additionally, model-motivated factors that are considered strong in [Chib et al. \(2022\)](#) display a similar pattern. However, Momentum and Return on Equity (RoE) remain strong across the sample period from 1967 to 2021 or from 2000 to 2021.

We apply the weighting BH FDR control procedure in [Genovese et al. \(2006\)](#) to the p-values obtained from the OLS and non-time-varying factor model. A back-of-the-envelope calculation shows that assigning weights that are more than 100 times higher is necessary to identify all economically important factors. However, in the recent sample from 2000 to 2021, this procedure can only increase the number of rejections among model-motivated factors by around 2, at the cost of a significant decrease in total rejection.

tions.

We propose a time-varying latent factor model, and find that the alphas display smooth changes over the sample period. This finding supports the use of a time-varying model and suggests the time-varying nature of the alpha for each factor. Moreover, in the case of without imposing asset pricing restrictions in the latent factor model, a generalized eigenvalue-ratio procedure reveals that 2 latent factors and the market factor can effectively account for almost all variations of all existing factors. In the case of imposing asset pricing restrictions for improved explanatory power, only one additional factor is needed. The results are largely consistent with the cluster analysis, suggesting no more than 4 factors in total, a number much smaller than most researchers believed

Our study adds to the existing literature on multiple testing in the asset pricing context by introducing a flexible time-varying latent factor framework. We establish the asymptotic properties of a local PCA-based estimator for the time-varying alpha and propose a generalized eigenvalue-ratio estimator for the latent dimension. Our proposed method is simple to implement and can be applied to other multiple testing and factor models in a number of areas, such as factor models for bonds, currencies, and mutual and hedge fund returns.

Table 1: BH procedure on full sample

The table reports the significant factors obtained via CAPM model in full sample:

$$r_{i,t} = \alpha_i + \beta_i f_t + u_{i,t},$$

where $r_{i,t}$ is the long-short strategy return of factors, and f_t is the market factor. The t statistic and p-value of α_i are shown in the table. The sample period is from 1967 to 2021. The dataset is constructed by [Chen and Zimmermann \(2021\)](#).

factor	t	p	factor	t	p
DivSeason	16.010	<0.001	EntMult	6.732	<0.001
AnnouncementReturn	15.093	<0.001	NetPayoutYield	6.651	<0.001
DelFINL	12.193	<0.001	hire	6.470	<0.001
DivYieldST	11.002	<0.001	MomSeason11YrPlus	6.456	<0.001
IndRetBig	10.971	<0.001	zerotrade	6.372	<0.001
dNoa	10.211	<0.001	zerotradeAlt12	6.341	<0.001
EarningsStreak	9.967	<0.001	MomOffSeason06YrPlus	6.229	<0.001
NumEarnIncrease	9.769	<0.001	BM	6.069	<0.001
NetDebtFinance	9.767	<0.001	MomSeason16YrPlus	6.000	<0.001
SmileSlope	9.903	<0.001	DownRecomm	6.042	<0.001
ShortInterest	9.526	<0.001	DelCOL	5.843	<0.001
ChTax	9.210	<0.001	roaq	5.811	<0.001
AnalystRevision	9.248	<0.001	Accruals	5.806	<0.001
InvestPPEInv	9.165	<0.001	VolSD	5.805	<0.001
ConvDebt	9.081	<0.001	PctAcc	5.710	<0.001
EarningsSurprise	8.898	<0.001	MomSeasonShort	5.708	<0.001
RevenueSurprise	8.747	<0.001	grcapx	5.694	<0.001
TrendFactor	8.745	<0.001	DebtIssuance	5.701	<0.001
STreversal	8.415	<0.001	PriceDelayRsq	5.681	<0.001
ShareIss1Y	8.315	<0.001	IntMom	5.661	<0.001
FirmAgeMom	8.148	<0.001	ExchSwitch	5.602	<0.001
ChInv	8.103	<0.001	ReturnSkew	5.592	<0.001
ResidualMomentum	7.861	<0.001	UpRecomm	5.632	<0.001
AssetGrowth	7.664	<0.001	SP	5.559	<0.001
Frontier	7.539	<0.001	CBOperProf	5.391	<0.001
zerotradeAlt1	7.532	<0.001	IdioRisk	5.379	<0.001
AccrualsBM	7.496	<0.001	std_turn	5.368	<0.001
VolumeTrend	7.483	<0.001	MS	5.361	<0.001
DelCOA	7.459	<0.001	ProbInformedTrading	5.453	<0.001
DelNetFin	7.438	<0.001	MomSeason06YrPlus	5.313	<0.001
BMdec	7.274	<0.001	ShareVol	5.307	<0.001
InvGrowth	7.211	<0.001	BetaFP	-5.254	<0.001
CompositeDebtIssuance	7.173	<0.001	ChEQ	5.247	<0.001
NOA	7.050	<0.001	MomVol	5.234	<0.001
VolMkt	7.017	<0.001	MaxRet	5.220	<0.001
ChangeInRecommendation	7.137	<0.001	ChInvIA	5.216	<0.001
ShareIss5Y	6.987	<0.001	Tax	5.210	<0.001
NetEquityFinance	6.920	<0.001	IdioVol3F	5.192	<0.001
XFIN	6.893	<0.001	grcapx3y	5.118	<0.001
retConglomerate	6.808	<0.001	RIO_Volatility	5.067	<0.001

Table 1 continued

factor	t	p	factor	alpha.t	alpha.p
FEPS	5.016	<0.001	EquityDuration	3.636	<0.001
MomSeason	4.936	<0.001	ShareRepurchase	3.616	<0.001
ReturnSkew3F	4.789	<0.001	IO_ShortInterest	3.613	<0.001
PctTotAcc	4.794	<0.001	AgeIPO	3.602	<0.001
DivInit	4.708	<0.001	Mom6m	3.517	<0.001
MomOffSeason	4.632	<0.001	CompEquIss	3.509	<0.001
DolVol	4.629	<0.001	GrAdExp	3.473	0.001
DelEqu	4.555	<0.001	Mom12m	3.454	0.001
IndIPO	4.540	<0.001	RIO_Turnover	3.422	0.001
ForecastDispersion	4.538	<0.001	AdExp	3.405	0.001
ChNNCOA	4.461	<0.001	PS	3.383	0.001
Mom12mOffSeason	4.451	<0.001	REV6	3.347	0.001
Recomm_ShortInterest	4.474	<0.001	DelLTI	3.327	0.001
MomRev	4.437	<0.001	IntanEP	3.309	0.001
OScore	4.426	<0.001	DelBreadth	3.268	0.001
OperProfRD	4.338	<0.001	EarnSupBig	3.256	0.001
AM	4.322	<0.001	Leverage	3.184	0.002
ChForecastAccrual	4.260	<0.001	sfe	3.174	0.002
skew1	4.280	<0.001	ChNWC	3.169	0.002
PriceDelayTstat	4.162	<0.001	PayoutYield	3.144	0.002
GrSaleToGrInv	4.140	<0.001	MomOffSeason16YrPlus	3.129	0.002
Illiquidity	4.133	<0.001	High52	3.044	0.002
CredRatDG	4.129	<0.001	PatentsRD	3.024	0.003
OperProf	4.065	<0.001	RIO_MB	2.983	0.003
PriceDelaySlope	3.966	<0.001	iomom_cust	2.965	0.003
RD	3.962	<0.001	cfp	2.943	0.003
RIO_Dis	3.948	<0.001	fgr5yrLag	2.940	0.003
EarningsForecastDisparity	3.863	<0.001	RoE	2.857	0.004
AOP	3.833	<0.001	EarningsConsistency	2.802	0.005
NetDebtPrice	3.824	<0.001	GP	2.784	0.006
IdioVolAHT	3.822	<0.001	tang	2.763	0.006
GrLTNOA	3.818	<0.001	MRreversal	2.731	0.006
OrgCap	3.810	<0.001	IntanCFP	2.726	0.007
Mom6mJunk	3.767	<0.001	RDS	2.727	0.007
RDIP	3.719	<0.001	CF	2.706	0.007
betaVIX	3.716	<0.001	DivOmit	2.685	0.007
MeanRankRevGrowth	3.698	<0.001	CoskewACX	2.684	0.007
CashProd	3.693	<0.001			
ChAssetTurnover	3.681	<0.001			
EBM	3.676	<0.001			

Table 2: BH procedure on recent sample

The table reports the results from the following regression:

$$r_{i,t} = \alpha_i + \beta_i f_t + u_{i,t},$$

where $r_{i,t}$ is the long-short strategy return of factors, and f_t is the market factor. The t statistic and p-value of α_i are shown in the table. In total, The sample period spans from 2000 to 2021. The dataset is constructed by [Chen and Zimmermann \(2021\)](#).

factor	t	p	factor	t	p
AnnouncementReturn	9.440	<0.001	ExchSwitch	3.732	<0.001
SmileSlope	8.152	<0.001	DolVol	3.667	<0.001
DivSeason	7.617	<0.001	NetEquityFinance	3.663	<0.001
ShortInterest	6.265	<0.001	RIO_Turnover	3.653	<0.001
VolumeTrend	6.040	<0.001	FEPS	3.634	<0.001
ConvDebt	5.657	<0.001	CompositeDebtIssuance	3.572	<0.001
NetDebtFinance	5.590	<0.001	UpRecomm	3.542	<0.001
EarningsStreak	5.525	<0.001	ShareIss5Y	3.522	0.001
ShareIss1Y	5.112	<0.001	dNoa	3.487	0.001
FirmAgeMom	5.018	<0.001	Recomm_ShortInterest	3.464	0.001
MomOffSeason06YrPlus	4.824	<0.001	EntMult	3.449	0.001
ChangeInRecommendation	4.728	<0.001	OrgCap	3.417	0.001
VolMkt	4.689	<0.001	skew1	3.363	0.001
NumEarnIncrease	4.604	<0.001	Tax	3.361	0.001
Frontier	4.355	<0.001	VolSD	3.346	0.001
OperProfRD	4.340	<0.001	roaq	3.296	0.001
RevenueSurprise	4.275	<0.001	zerotrade	3.270	0.001
CBOperProf	4.255	<0.001	BetaFP	-3.248	0.001
AccrualsBM	4.192	<0.001	RoE	3.231	0.001
IndRetBig	4.143	<0.001	SP	3.218	0.001
DivYieldST	4.131	<0.001	PriceDelayRsq	3.200	0.002
DelFINL	4.099	<0.001	MS	3.194	0.002
DownRecomm	4.018	<0.001	zerotradeAlt12	3.129	0.002
MomSeason16YrPlus	4.005	<0.001	AssetGrowth	3.106	0.002
NetPayoutYield	3.981	<0.001	Illiquidity	3.096	0.002
RIO_Volatility	3.965	<0.001	ChTax	3.085	0.002
XFIN	3.893	<0.001	OperProf	3.080	0.002
zerotradeAlt1	3.859	<0.001	IdioRisk	3.056	0.002

Table 3: The significant model-motivated factors

The table reports the results from the following regression:

$$r_{i,t} = \alpha_i + \beta_i^\top f_t + u_{i,t}, \quad 1 \leq i \leq N, 1 \leq t \leq T,$$

where $r_{i,t}$ is the long-short portfolio of model-motivated factors, and f_t is a vector of Fama-French 3 factors. The t value, p value as well as the BH rejection threshold are shown in the table. In total, there are 15 factors with the sampling period spanning from 2000 to 2021. The model-motivated factors are selected based on [Chib et al. \(2022\)](#).

Full sample			Recent Sample		
factor	t	p	factor	t	p
MomSeason11YrPlus	6.456	<0.001	MomOffSeason06YrPlus	4.824	<0.001
MomOffSeason06YrPlus	6.229	<0.001	MomSeason16YrPlus	4.005	<0.001
MomSeason16YrPlus	6.000	<0.001	RoE	3.231	0.001
MomSeasonShort	5.708	<0.001			
MomSeason06YrPlus	5.313	<0.001			
MomVol	5.234	<0.001			
MomSeason	4.936	<0.001			
MomOffSeason	4.632	<0.001			
Mom12mOffSeason	4.451	<0.001			
MomRev	4.437	<0.001			
Mom6m	3.517	<0.001			
Mom12m	3.454	0.001			
MomOffSeason16YrPlus	3.129	0.002			
RoE	2.857	0.004			

Table 4: GRW procedure on recent sample under pre-specified weights

The table reports the results of the GRW (or weighting BH) procedure on the p-values of model-motivated factors shown in Table 2. In the GRW procedure, each p-value is assigned a normalized weight and the adjusted p-value is the ratio of the original p-value on the weight. Then BH procedure is applied to the adjusted p-value. γ is the ratio of the weight on selected factors to the weight on benchmark factors.

factor	$\gamma = 2$ rejected	$\gamma = 5$ rejected	$\gamma = 10$ rejected
RoE	1	1	1
MomOffSeason06YrPlus	1	1	1
MomSeason16YrPlus	1	1	1
MomSeason	0	1	1
MomSeason06YrPlus	0	1	1
MomSeason11YrPlus	0	0	1
MomVol	0	0	1
Mom12mOffSeason	0	0	1
Mom6m	0	0	0
MomOffSeason16YrPlus	0	0	0
Mom12m	0	0	0
MomRev	0	0	0
MomOffSeason	0	0	0
Investment	0	0	0
MomSeasonShort	0	0	0

Table 5: BH rejections on full sample: factor model

The table reports the results of the alpha factor model on full sample spanning from 1967 to 2021. The factor model is given by:

$$r_{i,t} = \alpha_i + \beta_i^\top \lambda + \beta_i^\top (f_t - \mathbb{E}[f_t]) + u_{i,t}, \quad i \leq N, t \leq T$$

The estimates of α_i and its p-value are reported.

factor	p	factor	p	factor	p
IntanCFP	<0.001	MomSeason11YrPlus	<0.001	AccrualsBM	<0.001
ReturnSkew	<0.001	MomSeason06YrPlus	<0.001	dNoa	<0.001
zerotradeAlt1	<0.001	MomSeason	<0.001	DivYieldST	<0.001
IntanBM	<0.001	RDIPO	<0.001	XFIN	<0.001
IndRetBig	<0.001	MomOffSeason06YrPlus	<0.001	DivSeason	<0.001
ReturnSkew3F	<0.001	MomOffSeason	<0.001	DivOmit	<0.001
REV6	<0.001	Mom12mOffSeason	<0.001	DivInit	<0.001
IdioVol3F	<0.001	Mom12m	<0.001	DelNetFin	<0.001
IdioRisk	<0.001	LReversal	<0.001	DelLTI	<0.001
hire	<0.001	ResidualMomentum	<0.001	DelFINL	<0.001
RevenueSurprise	<0.001	IO_ShortInterest	<0.001	CompositeDebtIssuance	<0.001
RIO_Volatility	<0.001	InvGrowth	<0.001	skew1	<0.001
roaq	<0.001	EarningsSurprise	<0.001	DelEqu	<0.001
GrSaleToGrInv	<0.001	EarningsStreak	<0.001	SmileSlope	<0.001
IntanSP	<0.001	ShareRepurchase	<0.001	DebtIssuance	<0.001
RoE	<0.001	PctTotAcc	<0.001	ConvDebt	<0.001
grcapx	<0.001	ChTax	<0.001	DelCOA	<0.001
GP	<0.001	AnnouncementReturn	<0.001	VolumeTrend	<0.001
sfe	<0.001	ChNNCOA	<0.001	DelCOL	<0.001
Frontier	<0.001	STreversal	<0.001	Mom6mJunk	0.002
ShareIss1Y	<0.001	ChInvIA	<0.001	EarningsConsistency	0.002
ForecastDispersion	<0.001	ChInv	<0.001	RIO_MB	0.002
FirmAgeMom	<0.001	ChForecastAccrual	<0.001	TotalAccruals	0.002
ShareIss5Y	<0.001	ChEQ	<0.001	Recomm_ShortInterest	0.002
PctAcc	<0.001	ChangeInRecommendation	<0.001	ShareVol	0.002
FEPS	<0.001	cfp	<0.001	CredRatDG	0.002
ExchSwitch	<0.001	SurpriseRD	<0.001	betaVIX	0.002
ShortInterest	<0.001	CBOperProf	<0.001	CoskewACX	0.002
EntMult	<0.001	CashProd	<0.001	AgeIPO	0.002
grcapx3y	<0.001	Cash	<0.001	MS	0.002
IntMom	<0.001	tang	<0.001	MaxRet	0.003
retConglomerate	<0.001	Tax	<0.001	Mom6m	0.003
InvestPPEInv	<0.001	AM	<0.001	IndIPO	0.003
Price	<0.001	AnalystRevision	<0.001	Leverage	0.004
ProbInformedTrading	<0.001	BMdec	<0.001	Coskewness	0.004
OScore	<0.001	BM	<0.001	OperProfRD	0.004
RD	<0.001	BidAskSpread	<0.001	RIO_Disp	0.004
OptionVolume1	<0.001	BetaTailRisk	<0.001	VolMkt	0.004
OPLeverage	<0.001	TrendFactor	<0.001	OrderBacklogChg	0.005
OperProf	<0.001	UpRecomm	<0.001	ChAssetTurnover	0.005
NumEarnIncrease	<0.001	Beta	<0.001	RDS	0.005
NOA	<0.001	AssetGrowth	<0.001	EquityDuration	0.005
NetPayoutYield	<0.001	VarCF	<0.001	sinAlgo	0.005
NetEquityFinance	<0.001	SP	<0.001	MomOffSeason16YrPlus	0.006
NetDebtPrice	<0.001	CompEquIss	<0.001	IdioVolAHT	0.006
NetDebtFinance	<0.001	ChNWC	<0.001	GrAdExp	0.006
RDcap	<0.001	AdExp	<0.001	EarningsForecastDisparity	0.006
MomVol	<0.001	DownRecomm	<0.001	OrgCap	0.007
MomSeasonShort	<0.001	DolVol	<0.001	EBM	0.007
MomSeason16YrPlus	<0.001	Accruals	<0.001	MomRev	0.007

Table 6: BH rejections on recent sample: factor model

The table reports the results of the alpha factor model on recent sample spanning from 2000 to 2021. The factor model is given by:

$$r_{i,t} = \alpha_i + \beta_i^\top \lambda + \beta_i^\top (f_t - \mathbb{E}[f_t]) + u_{i,t}, \quad i \leq N, t \leq T$$

The estimates of α_i and its p-value are reported.

factor	p	factor	p
CBOperProf	0.001	SmileSlope	0.001
BM	0.001	OptionVolume1	0.001
NetEquityFinance	0.001	ExchSwitch	0.001
ConvDebt	0.001	NumEarnIncrease	0.001
NetDebtFinance	0.001	XFIN	0.001
RevenueSurprise	0.001	RoE	0.001
Cash	0.001	AnnouncementReturn	0.001
NetPayoutYield	0.001	VolumeTrend	0.001
FEPS	0.001	roaq	0.001
UpRecomm	0.001	Frontier	0.001
MomSeason16YrPlus	0.001	EarningsStreak	0.001
ChangeInRecommendation	0.001	RIO_Volatility	0.002
DownRecomm	0.001	IndRetBig	0.002
CompositeDebtIssuance	0.001	Tax	0.002
SP	0.001	MomOffSeason06YrPlus	0.002
DelFINL	0.001	dNoa	0.002
ShortInterest	0.001	OPLeverage	0.002
DivSeason	0.001	MS	0.002
FirmAgeMom	0.001	NetDebtPrice	0.002
RD	0.001	skew1	0.002
AccrualsBM	0.001	OperProf	0.002
ShareIss1Y	0.001	AssetGrowth	0.002

Table 7: BH rejections of model-motivated factors on full and recent sample: factor model

The table reports the results of the alpha factor model on model-motivated factors. The factor model is given by:

$$r_{i,t} = \alpha_i + \beta_i^\top \lambda + \beta_i^\top (f_t - \mathbb{E}[f_t]) + u_{i,t}, \quad i \leq N, t \leq T$$

The estimates of α_i and its p-value are reported.

Full sample		Recent sample	
factor	p-value	factor	p-value
RoE	0.001	MomSeason16YrPlus	0.001
MomVol	0.001	RoE	0.001
MomSeasonShort	0.001	MomOffSeason06YrPlus	0.002
MomSeason16YrPlus	0.001		
MomSeason11YrPlus	0.001		
MomSeason06YrPlus	0.001		
MomSeason	0.001		
MomOffSeason06YrPlus	0.001		
MomOffSeason	0.001		
Mom12mOffSeason	0.001		
Mom12m	0.001		
Mom6m	0.003		
MomOffSeason16YrPlus	0.006		
MomRev	0.007		

Table 8: Cluster assignments of factors identified via CAPM in full sample

The table presents the clustering results of factors identified via CAPM in full sample. 5 clusters chosen based on Figure 8.

Cluster 1	Cluster 2	Cluster 3	Cluster 4	Cluster 5
Accruals	AnalystRevision	AssetGrowth	BetaFP	AgeIPO
AccrualsBM	AnnouncementReturn	ChAssetTurnover	CBOperProf	AOP
AdExp	ChangeInRecommendation	ChEQ	ConvDebt	betaVIX
AM	ChForecastAccrual	ChInv	GP	CF
BM	ChTax	ChInvIA	IdioRisk	cfp
BMdec	CompEquIss	ChNNCOA	IdioVol3F	DelLTI
CashProd	CoskewACX	CompositeDebtIssuance	IdioVolAHT	DivInit
ChNWC	CredRatDG	DebtIssuance	MaxRet	DivOmit
DivSeason	DelBreadth	DelCOA	OperProfRD	FEPS
DivYieldST	DownRecomm	DelCOL	OrgCap	fgr5yrLag
EBM	EarningsForecastDisparity	DelEqu	PriceDelayTstat	ForecastDispersion
EntMult	EarningsStreak	DelFINL	ProbInformedTrading	IndIPO
EquityDuration	EarningsSurprise	DelNetFin	PS	NetEquityFinance
Frontier	EarnSupBig	dNoa	Recomm_ShortInterest	NetPayoutYield
GrLTNOA	FirmAgeMom	DoVol	ShareVol	OperProf
IntanCFP	High52	EarningsConsistency	ShortInterest	OScore
IntanEP	IndRetBig	ExchSwitch	SmileSlope	RD
Leverage	IntMom	GrAdExp	std_turn	ReturnSkew
NetDebtPrice	iomom_cust	grcapx	VolMkt	ReturnSkew3F
PayoutYield	Mom12m	grcapx3y	VolSD	roaq
PctAcc	Mom12mOffSeason	GrSaleToGrInv	zerotrade	RoE
RIO_Dis	Mom6m	hire	zerotradeAlt1	sfe
RIO_MB	Mom6mJunk	Illiquidity	zerotradeAlt12	ShareIss1Y
RIO_Turnover	MomRev	InvestPPEInv		ShareRepurchase
RIO_Volatility	MomSeasonShort	InvGrowth		tang
SP	MomVol	IO_ShortInterest		Tax
	NumEarnIncrease	MeanRankRevGrowth		XFIN
	ResidualMomentum	MomOffSeason		
	retConglomerate	MomOffSeason06YrPlus		
	REV6	MomOffSeason16YrPlus		
	RevenueSurprise	MomSeason		
	skew1	MomSeason06YrPlus		
	STreversal	MomSeason11YrPlus		
	TrendFactor	MomSeason16YrPlus		
	UpRecomm	MRreversal		
		MS		
		NetDebtFinance		
		NOA		
		PatentsRD		
		PctTotAcc		
		PriceDelayRsqr		
		PriceDelaySlope		
		RDIPO		
		RDS		
		ShareIss5Y		
		VolumeTrend		

Table 9: Cluster assignments of factors identified via CAPM in recent sample

The table presents the assignment of factors identified via CAPM in recent sample. 3 clusters are chosen based on Figure 9.

Cluster 1	Cluster 2	Cluster 3
AnnouncementReturn	DivSeason	ConvDebt
SmileSlope	EarningsStreak	VolMkt
ShortInterest	ShareIss1Y	OperProfRD
VolumeTrend	NumEarnIncrease	CBOperProf
NetDebtFinance	Frontier	zerotradeAlt1
FirmAgeMom	RevenueSurprise	Recomm_ShortInterest
MomOffSeason06YrPlus	AccrualsBM	VolSD
ChangeInRecommendation	DivYieldST	zerotrade
IndRetBig	MomSeason16YrPlus	BetaFP
DelFINL	NetPayoutYield	MS
DownRecomm	XFIN	zerotradeAlt12
RIO_Volatility	NetEquityFinance	IdioRisk
ExchSwitch	FEPS	
DolVol	EntMult	
RIO_Turnover	Tax	
CompositeDebtIssuance	roaq	
UpRecomm	RoE	
ShareIss5Y	SP	
dNoa	ChTax	
OrgCap	OperProf	
skew1		
PriceDelayRsq		
AssetGrowth		
Illiquidity		

Table 10: Cluster assignments of model-motivated factors

The table presents the assignment of model-motivated factors identified using CAPM when three clusters are chosen in full sample and in recent sample.

Full sample		
Cluster 2	Cluster 3	Cluster 5
MomSeasonShort	MomSeason11YrPlus	RoE
MomVol	MomOffSeason06YrPlus	
Mom12mOffSeason	MomSeason16YrPlus	
MomRev	MomSeason06YrPlus	
Mom6m	MomSeason	
Mom12m	MomOffSeason	
	MomOffSeason16YrPlus	
Recent sample		
Cluster 1	Cluster 2	
MomOffSeason06YrPlus	MomSeason16YrPlus	
	RoE	

Table 11: Factors with significance decay after publication

The table reports factors with significance decay, i.e. t-stat decrease, after publication. The CAPM for each factor with sample before and after publication is implemented. The samples before and after the publication are of the same size. The t-stats of α are compared. In total, 157 factors experience decline in t-stat after publication. Among them, 72 of them drop from above 1.96 to below 1.96, the critical value of t test of 5 percent size.

Data source: [Chen and Zimmermann \(2021\)](#).

factors	year of publication	t-stat drop	<1.96 after publication	factors	year of publication	t-stat drop	<1.96 after publication
AbnormalAccruals	2001	0.48	Yes	BMdec	1992	1.25	No
Accruals	1996	6.74	Yes	BPEBM	2007	0.33	No
AdExp	2001	2.34	Yes	BrandInvest	2014	0.08	No
AnalystValue	1998	0.88	Yes	Cash	2012	0.38	No
AOP	1998	3.28	Yes	CashProd	2009	0.72	No
betaVIX	2006	3.14	Yes	ChangelnRecommendation	2004	5.36	No
BookLeverage	1992	3.18	Yes	ChInv	2002	4.09	No
CF	1994	1.73	Yes	ChNAnalyst	2008	1.72	No
ChAssetTurnover	2008	1.88	Yes	ChNWC	2008	0.19	No
ChEQ	2010	1.99	Yes	CompositeDebtIssuance	2008	2.13	No
ChForecastAccrual	2004	1.18	Yes	ConvDebt	2016	1.74	No
ChInvIA	1998	3.73	Yes	CredRatDG	2001	1.55	No
ChNNCOA	2008	3.39	Yes	CustomerMomentum	2008	0.23	No
ConsRecomm	2002	0.70	Yes	DebtIssuance	1999	4.88	No
CoskewACX	2006	1.75	Yes	DelDRC	2012	0.72	No
Coskewness	2000	2.21	Yes	DelFINL	2005	7.41	No
DelBreadth	2002	0.60	Yes	DivInit	1995	3.99	No
DelCOA	2005	5.98	Yes	DownRecomm	2002	1.93	No
DelCOL	2005	5.81	Yes	EarningsStreak	2012	0.38	No
DelEqu	2005	3.25	Yes	EarningsSurprise	1984	4.17	No
DelLTI	2005	1.96	Yes	EBM	2007	1.27	No
DelNetFin	2005	2.62	Yes	ExchSwitch	1995	0.04	No
DivSeason	2013	4.73	Yes	FEPS	2006	1.16	No
dNoa	2004	7.15	Yes	FirmAgeMom	2004	0.59	No
EarningsForecastDisparity	2011	1.17	Yes	ForecastDispersion	2002	0.51	No
EntMult	2011	3.74	Yes	Frontier	2009	1.97	No
EP	1977	1.35	Yes	GrAdExp	2014	0.57	No
EquityDuration	2004	2.93	Yes	hire	2014	1.44	No
fgr5yrLag	1996	1.95	Yes	IdioRisk	2006	0.64	No
grcapx	2006	4.32	Yes	IdioVol3F	2006	0.38	No
grcapx3y	2006	4.40	Yes	IndRetBig	2007	1.82	No
GrLTNOA	2003	2.00	Yes	IntanBM	2006	0.93	No
GrSaleToGrInv	1998	4.09	Yes	IntanCFP	2006	1.66	No
Herf	2006	2.44	Yes	Investment	2004	0.02	No
HerfAsset	2006	1.53	Yes	InvestPPEInv	2008	2.24	No
HerfBE	2006	1.95	Yes	InvGrowth	2012	0.29	No
IdioVolAHT	2003	1.64	Yes	IO_ShortInterest	2005	1.02	No
IntanEP	2006	2.06	Yes	Leverage	1988	2.04	No
IntanSP	2006	1.75	Yes	Mom12m	1993	2.04	No
IntMom	2012	1.84	Yes	Mom6m	1993	0.26	No
LRreversal	1985	0.43	Yes	MomOffSeason06YrPlus	2008	1.71	No
MeanRankRevGrowth	1994	4.56	Yes	MomSeason16YrPlus	2008	0.44	No
Mom12mOffSeason	2008	1.22	Yes	MomVol	2000	4.20	No
Mom6mJunk	2007	1.34	Yes	MS	2005	0.75	No
MomOffSeason	2008	3.25	Yes	NetDebtFinance	2006	3.96	No
MomOffSeason11YrPlus	2008	1.96	Yes	NetEquityFinance	2006	0.28	No
MomRev	2006	2.89	Yes	NetPayoutYield	2007	0.39	No
MomSeason	2008	5.15	Yes	NumEarnIncrease	2012	0.35	No
MomSeason06YrPlus	2008	3.13	Yes	OperProf	2006	0.69	No
MomSeason11YrPlus	2008	1.55	Yes	OperProfRD	2016	0.51	No
MomSeasonShort	2008	2.25	Yes	OptionVolume2	2012	0.51	No
NOA	2004	5.88	Yes	OScore	1998	1.32	No
OPLeverage	2010	2.01	Yes	Price	1972	0.72	No
OptionVolume1	2012	3.14	Yes	PS	2000	1.15	No
OrderBacklog	2003	1.34	Yes	RDcap	2011	0.51	No
OrgCap	2013	1.30	Yes	RDIPO	2006	0.52	No
PayoutYield	2007	2.25	Yes	RDS	2011	0.90	No
PctAcc	2011	2.28	Yes	retConglomerate	2012	0.03	No
PredictedFE	1998	1.71	Yes	ReturnSkew3F	2015	0.66	No
PriceDelayRsqr	2005	2.74	Yes	RevenueSurprise	2006	2.28	No
PriceDelaySlope	2005	1.53	Yes	RIO_Volatility	2005	1.51	No
PriceDelayTstat	2005	2.29	Yes	ShareIss1Y	2008	2.53	No
RD	2001	2.73	Yes	ShareIss5Y	2006	2.10	No
Recomm_ShortInterest	2011	1.68	Yes	ShareRepurchase	1995	1.34	No
REV6	1996	4.52	Yes	ShareVol	1998	2.13	No
RIO_Disb	2005	0.87	Yes	ShortInterest	2001	1.30	No
RIO_MB	2005	2.62	Yes	sinAlgo	2009	1.67	No
RIO_Turnover	2005	1.38	Yes	SmileSlope	2011	1.19	No
Size	1981	1.13	Yes	SP	1996	1.07	No
SurpriseRD	2004	1.89	Yes	Spinoff	1993	1.48	No
TotalAccruals	2005	1.87	Yes	std_turn	2001	3.54	No
VarCF	1996	1.74	Yes	STreversal	1989	8.37	No
AccrualsBM	2004	3.35	No	tang	2009	1.05	No
Activism2	2005	0.16	No	Tax	2004	0.28	No
AgeIPO	1991	1.97	No	UpRecomm	2002	4.16	No
AM	1992	1.02	No	VolMkt	1996	0.55	No
AnnouncementReturn	1996	0.20	No	XFIN	2006	0.85	No
AssetGrowth	2008	2.03	No	zerotradeAlt1	2006	1.07	No
BetaLiquidityPS	2003	1.56	No				
Total			157				
Significance downgrade			72				

Table 12: Factors being identified under BH procedure before and after publication

The table presents the factors that pass the BH FDR control procedure before and after its publication date. The p values of each factor before and after publication is obtained from results of Table 11. The false discovery rate is controlled at 1 percent.

Before Publication		After Publication
STreversal	OScore	AnnouncementReturn
AnnouncementReturn	Frontier	DivYieldST
DelFINL	AssetGrowth	AnalystRevision
EarningsSurprise	Mom12m	VolumeTrend
dNoa	AM	ShortInterest
ChangeInRecommendation	ExchSwitch	SmileSlope
ShortInterest	BookLeverage	VolMkt
DebtIssuance	XFIN	BMdec
NetDebtFinance	DelEqu	EarningsSurprise
Accruals	CredRatDG	BM
UpRecomm	RIO_Volatility	FirmAgeMom
DelCOA	EntMult	EarningsStreak
AnalystRevision	ShareRepurchase	ExchSwitch
NOA	VolSD	VolSD
ChInv	Tax	zerotradeAlt1
MomVol	RD	STreversal
SmileSlope	IO_ShortInterest	IndIPO
AccrualsBM	fgr5yrLag	zerotrade
DivInit	ChNNCOA	DolVol
ShareIss1Y	Recomm_ShortInterest	IndRetBig
DivSeason	MomOffSeason	BetaFP
BMdec	ForecastDispersion	ShareIss1Y
DelCOL	zerotradeAlt12	DownRecomm
std_turn	NetPayoutYield	zerotradeAlt12
MomSeason	MomRev	MaxRet
REV6	NumEarnIncrease	Tax
ChInvIA	EquityDuration	RoE
VolMkt	FEPS	SP
DownRecomm	PriceDelayTstat	NetDebtFinance
IndRetBig	IdioRisk	MS
RevenueSurprise	IdioVolAHT	
grcapx	DolVol	
FirmAgeMom	PS	
zerotradeAlt1	EP	
DivYieldST	NetEquityFinance	
grcapx3y	OptionVolume1	
InvestPPEInv	MomSeason11YrPlus	
GrSaleToGrInv	MomSeason16YrPlus	
EarningsStreak	BM	
ShareIss5Y	OperProf	
MeanRankRevGrowth	Herf	
AOP	ChEQ	
CompositeDebtIssuance	RIO_Turnover	
MomOffSeason06YrPlus	IdioVol3F	
SP	zerotrade	
ConvDebt	Mom6m	
AgeIPO	PriceDelaySlope	
VolumeTrend	Mom12mOffSeason	
MomSeason06YrPlus	RDIP0	
Leverage	HerfBE	
DelNetFin	Illiquidity	
AdExp	OperProfRD	
ShareVol	IndIPO	
PriceDelayRsqr	IntMom	
betaVIX	ChForecastAccrual	
MS	CF	
Total	112	30

Table 13: Result table of post-publication and out-of-sample regression

Table 13 reports the results of the following regression:

$$R_{i,t} = \alpha_i + \beta_1 * \text{Out-of-sample}_{i,t} + \beta_2 * \text{Post-publication}_{i,t} + \epsilon_{i,t}$$

where

- $R_{i,t}$ is the long-short strategy return of factor i in period t .
- $\text{Out-of-sample}_{i,t}$ is a dummy equal to 1 when t is after the sampling period of the original paper of factor i .
- $\text{Post-publication}_{i,t}$ is a dummy equal to 1 when t is after the publication of factor i .

The results provide evidence that factors tend to lose power after publication.

Variable	(1)	(2)	(3)
Out of sample	-0.28*** (0.02)		-0.28*** (0.02)
Post-Publication		-0.3*** (0.03)	-0.3*** (0.03)
Constant	0.66*** (0.01)	0.65*** (0.01)	0.65*** (0.01)
Observations	126463	126463	126463

Table 14: Comparison among RP-PCA, FF3, and FF5

The table shows the anomalies with significant alphas from the following regressions:

$$r_{i,t} = \alpha_i + \beta_i f_t + u_{i,t},$$

where f_t is 3 RP-PCA latent factors and market factor, Fama-French 3 factors and Fama-French 5 factors respectively.

3 RP-PCA latents + market			Fama-French 3			Fama-French 5		
Anomaly	alpha	p-value	Anomaly	alpha	p-value	Anomaly	alpha	p-value
ConvDebt	0.4560	< 0.0001	DivSeason	0.2585	< 0.0001	DivSeason	0.2504	< 0.0001
DivSeason	0.2642	< 0.0001	VolumeTrend	0.7903	< 0.0001	NetDebtPrice	1.2205	< 0.0001
DivYieldST	0.5749	< 0.0001	VolMkt	1.0182	< 0.0001	ConvDebt	0.3953	< 0.0001
EarningsConsistency	0.4015	< 0.0001	ConvDebt	0.5010	< 0.0001	MomOffSeason06YrPlus	0.9380	< 0.0001
ExchSwitch	1.0259	< 0.0001	ShareIss1Y	0.9382	< 0.0001	VolumeTrend	0.5633	< 0.0001
Frontier	1.0141	< 0.0001	OperProfRD	1.1837	< 0.0001	Frontier	1.3895	< 0.0001
IndRetBig	1.2613	< 0.0001	CBOperProf	1.1013	< 0.0001	RD	1.1864	< 0.0001
MomOffSeason06YrPlus	0.9042	< 0.0001	RoE	0.6492	< 0.0001	DivYieldST	0.5093	0.0001
MomSeason16YrPlus	0.6569	< 0.0001	NumEarnIncrease	0.4215	< 0.0001	ExchSwitch	1.0765	0.0001
NetPayoutYield	0.9307	< 0.0001	NetPayoutYield	1.3125	< 0.0001	AccrualsBM	1.0928	0.0001
ShareIss1Y	0.6278	< 0.0001	zerotradeAlt1	0.9601	< 0.0001	zerotradeAlt1	0.8128	0.0002
			MomOffSeason06YrPlus	0.9398	< 0.0001	RevenueSurprise	0.4801	0.0002
			RevenueSurprise	0.5576	< 0.0001	dNoa	0.4263	0.0002
			AccrualsBM	1.1777	< 0.0001	zerotradeAlt1 2	0.5940	0.0003
			Frontier	1.4096	< 0.0001	NOA	0.7833	0.0003
			OperProf	0.8359	< 0.0001	MomSeason16YrPlus	0.6171	0.0004
			EntMult	0.7521	< 0.0001	ShareIss1Y	0.5509	0.0005
			roaq	1.2569	< 0.0001	IndRetBig	1.1104	0.0005
			Tax	0.5006	< 0.0001	OperProfRD	0.6962	0.0006
			BetaFP	-1.3434	< 0.0001	AssetGrowth	0.7759	0.0007
			SP	0.8535	< 0.0001	CompositeDebtIssuance	0.2441	0.0007
			VolSD	0.5688	< 0.0001	DelFINL	0.2788	0.0008
			IdioRisk	1.2535	< 0.0001	NumEarnIncrease	0.2895	0.0008
			IdioVol3F	1.2234	0.0001	CBOperProf	0.7008	0.0008
			zerotradeAlt1 2	0.6251	0.0001	VolSD	0.4782	0.0009
			IndRetBig	1.2913	0.0001	DolVol	0.6736	0.0011
			zerotrade	0.8546	0.0001	VolMkt	0.5038	0.0011
			DivYieldST	0.4793	0.0001	zerotrade	0.7282	0.0012
			MomSeason16YrPlus	0.6484	0.0001			
			DelFINL	0.3081	0.0001			
			RIO_Volatility	1.0383	0.0001			
			MaxRet	1.2355	0.0002			
			BMdec	0.4967	0.0002			
			IdioVolAHT	1.1259	0.0003			
			ExchSwitch	0.9600	0.0003			
			RIO_Turnover	0.8127	0.0004			
			DolVol	0.6863	0.0005			
			CompositeDebtIssuance	0.2358	0.0005			
			ChTax	0.4405	0.0007			
			std_turn	0.9718	0.0008			
			ShareIss5Y	0.4311	0.0009			
			dNoa	0.4473	0.0009			
			EarningsConsistency	0.4074	0.0013			
			OrgCap	0.5381	0.0013			
			NetDebtPrice	0.7854	0.0017			
			Illiquidity	0.3230	0.0022			
			AssetGrowth	0.7621	0.0024			

Table 15: Comparison among RP-PCA, FF3, and FF5, continued

The table compares the root means square, R^2 , and the number of significant anomalies after controlling for FDR among three models specified in the last table.

Factors	RMS of alpha	# Rejections	R Squared
3 latents + Mkt	0.397	11	0.574
FF3	0.608	47	0.370
FF5	0.462	28	0.438

Table 16: Fama-French factors on RP-PCA factors and vice versa

The table shows the significant alphas from the regressions

$$y_{i,t} = a_i + \beta_i x_t + u_{i,t}$$

where (y_t, x_t) is FF5, RP-PCA latents and Market (first panel) and RP-PCA latents and Market, FF5 (second panel), respectively.

FF5 on 3 Latents + Mkt					
factor	alpha	t-stat	p-value	fdr reject	p < 0.05
SMB	0.370	2.289	0.023	No	Yes
HML	-0.233	-1.157	0.248	No	No
RMW	0.232	1.927	0.055	No	No
CMA	0.013	0.110	0.912	No	No

3 latents on FF3					
factor	alpha	t-stat	p-value	fdr reject	p < 0.05
latent 1	0.394	3.061	0.002	Yes	Yes
latent 2	0.052	0.495	0.621	No	No
latent 3	-0.363	-4.253	<0.001	Yes	Yes

Table 17: Sharpe ratio of optimal portfolios

The table shows the Sharpe ratio of the optimal factors constructed by 3 RP-PCA latent factors, FF3 factors, and FF5 factors, respectively.

factors	Sharpe ratio
3 latents + Mkt	0.420
FF3	0.173
FF5	0.393

Figure 1: The rolling number of identified factors

The figure reports the rolling number of identified factors in the rolling study in Section 3.2. The rolling window is 20 years. Factors with missing data points in a rolling period are removed.

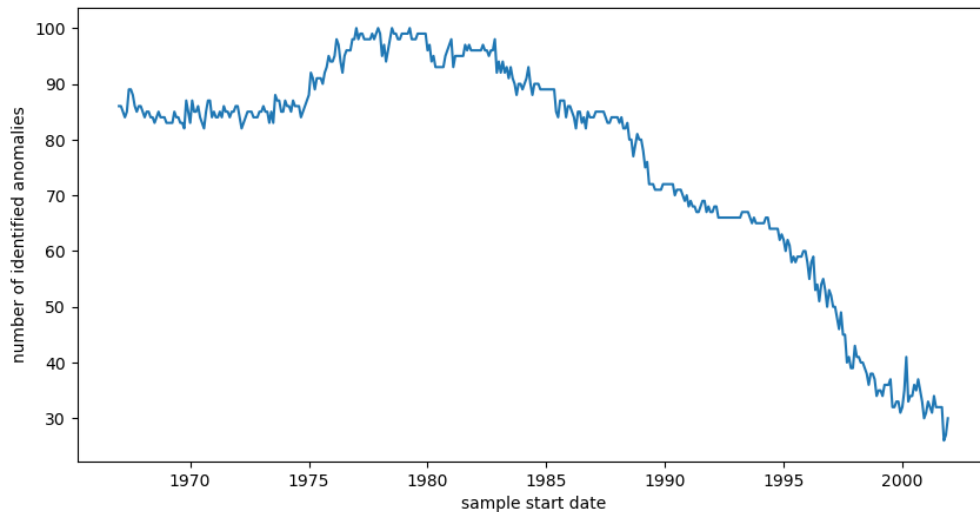


Figure 2: Rolling BH procedure

The figure reports the rolling number of identified factors across rolling samples. The detail of the rolling study design is in Section 3.2.

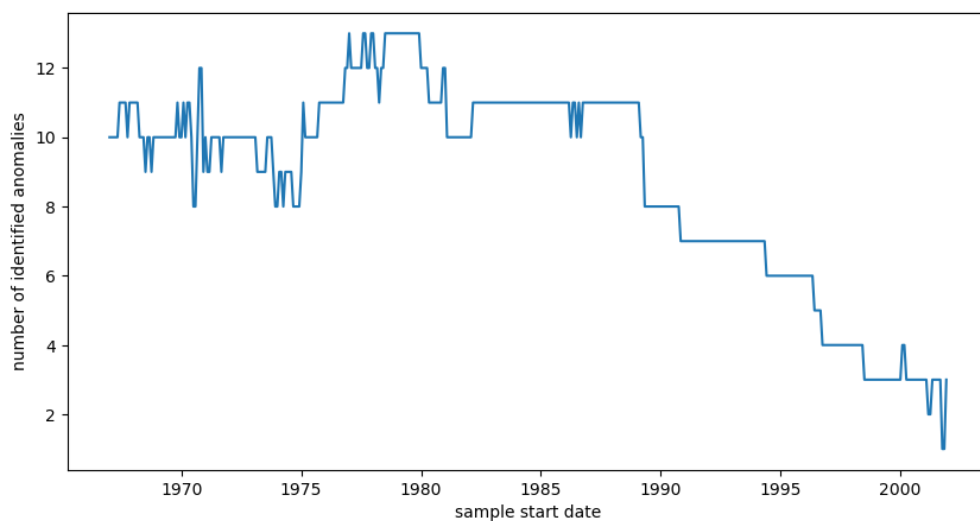


Figure 3: Rolling number of rejections: factor model

The figure presents the results of the rolling study using constant-loading factor model with the mix of market factor and unknown latent factors. The rolling window is 20 years.

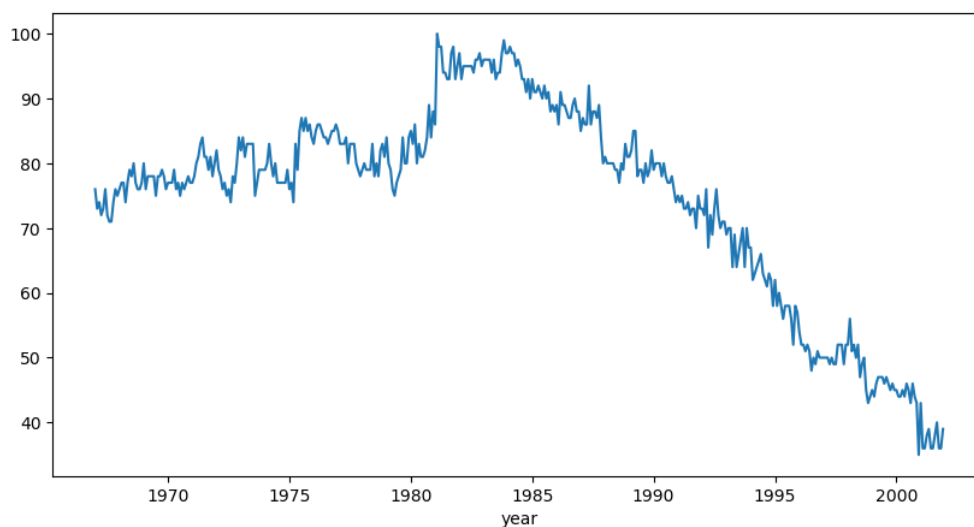


Figure 4: Alpha for the factor price: time-varying factor model

The figure presents the heat map of alpha estimates via time-varying factor model on full sample spanning from 1967 to 2021. The detail of the model is discussed in 2.4. The color density shows the value of the alpha.



Figure 5: Alpha for the model-motivated factor price: time-varying factor model

The figure presents the heat map of alpha estimates of the model-motivated factors via time-varying factor model on full sample spanning from 1967 to 2021. The detail of the model is discussed in 2.4. The color density shows the value of the alpha.

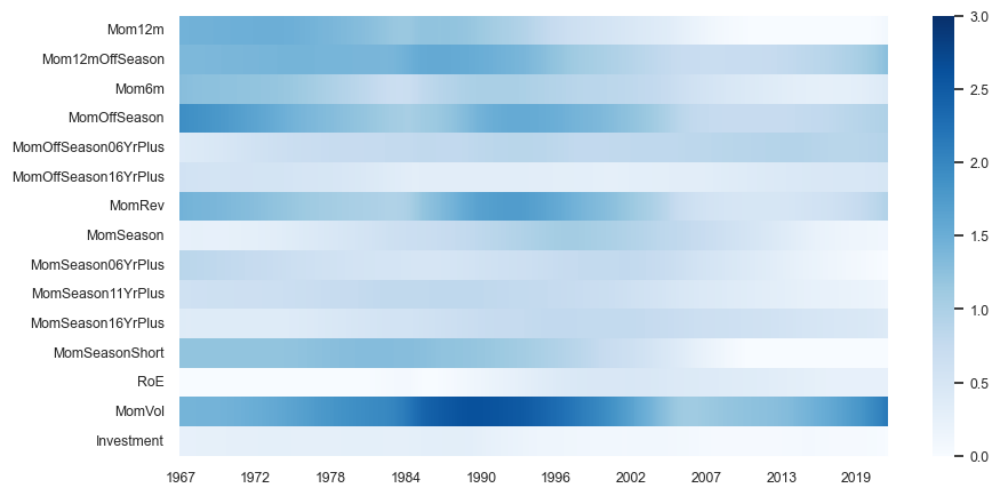


Figure 6: Number of BH rejections across time: time-varying model

The figure shows the number of identified factors after implementing the BH procedure on the p-value on alphas obtained from the time-varying model. The sample period spans from 1967 to 2021.

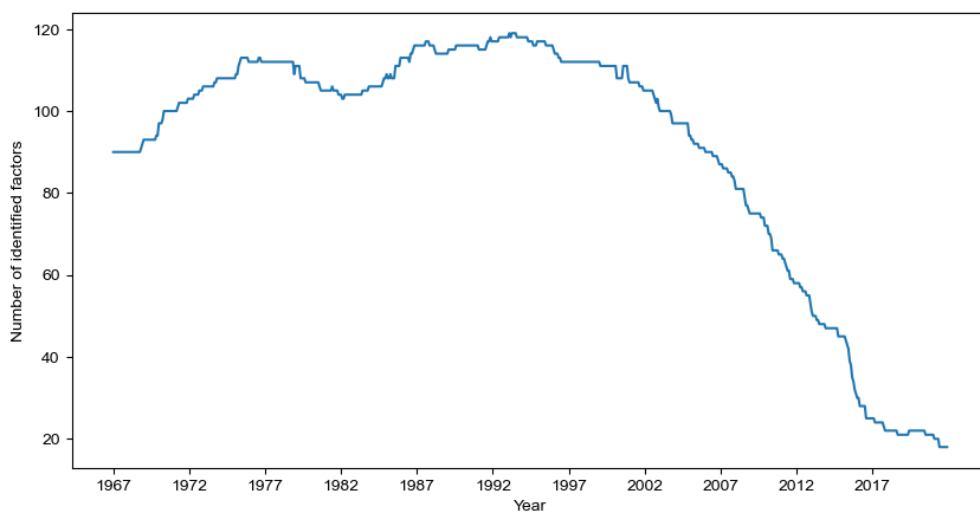


Figure 7: Number of BH rejections of model-motivated factors across time: time-varying model

The figure shows the number of model-motivated factors identified after implementing the BH procedure on the p-value on alphas obtained from the time-varying model. The sample period spans from 1967 to 2021.

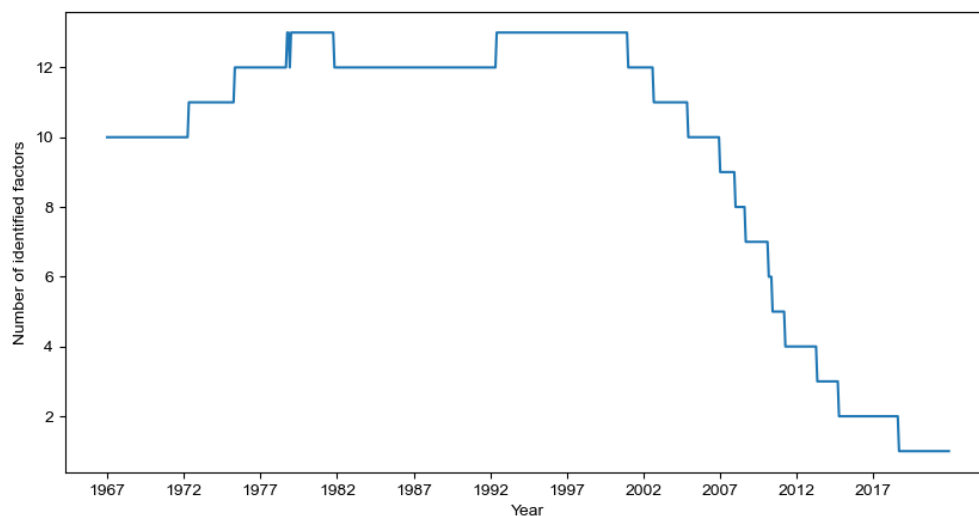


Figure 8: The aggregation levels across number of clusters: full sample

The figure plots the aggregation levels of different number of clusters of factors identified via CAPM in Section 1. The height on y-axis shows the proportion of the intergroup dissimilarity. According to this plot, 5 clusters are chosen that groups identified factors.

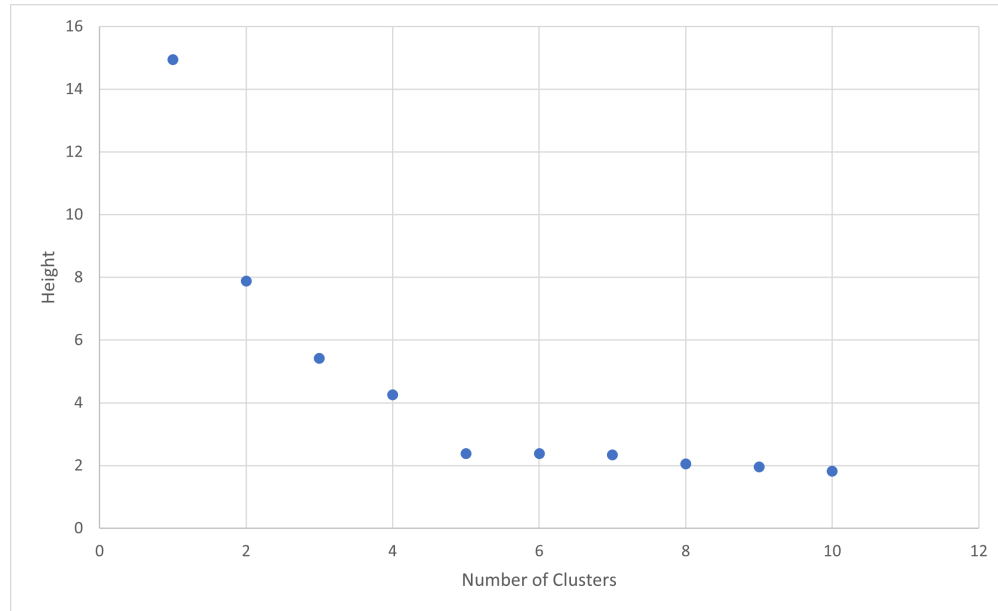


Figure 9: The aggregation levels across number of clusters: recent sample

The figure plots the aggregation levels of different number of clusters of factors identified via CAPM in Section 2. The height on y-axis shows the proportion of the intergroup dissimilarity. According to this plot, 3 clusters are chosen that groups identified factors.

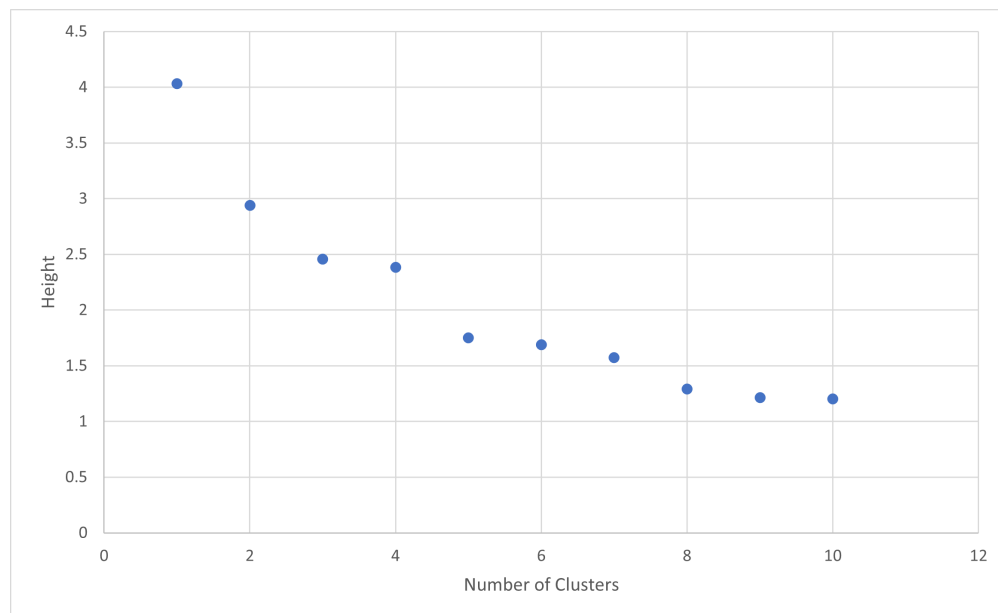


Figure 10: Correlation heat map of factors identified via CAPM in full sample
The heat map shows the correlation between factors identified via CAPM discussed in Section 2.1. The tail number of each factor indicates the cluster number that the factor belongs to. And the diagonal entries are replaced with 0 for better visualization effect.

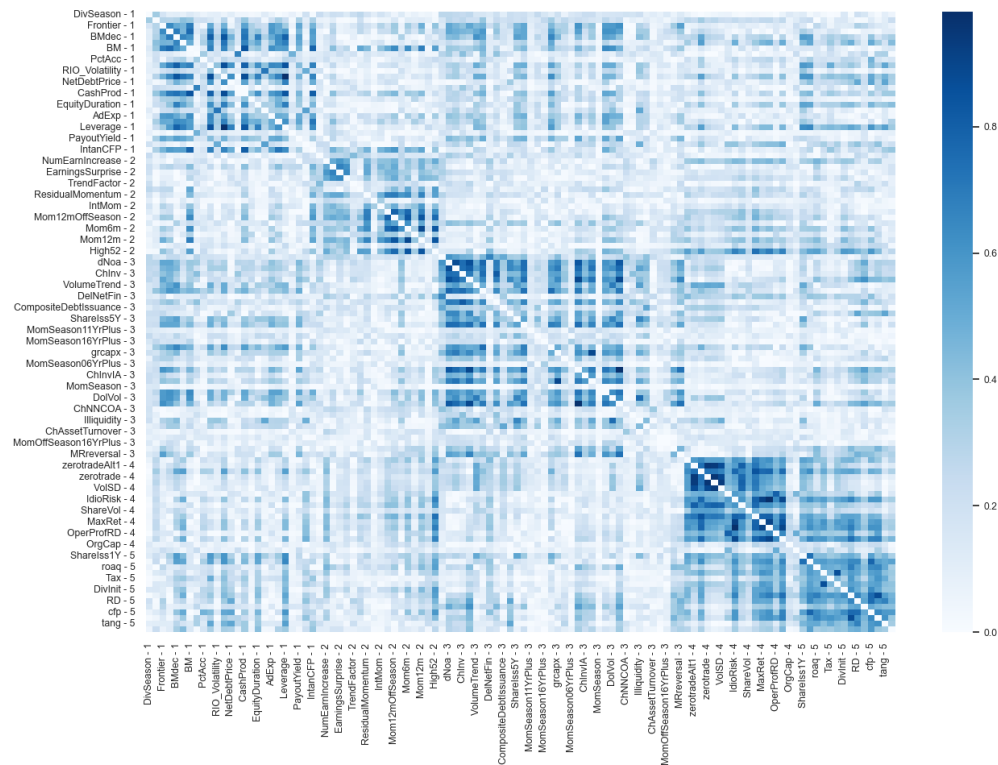
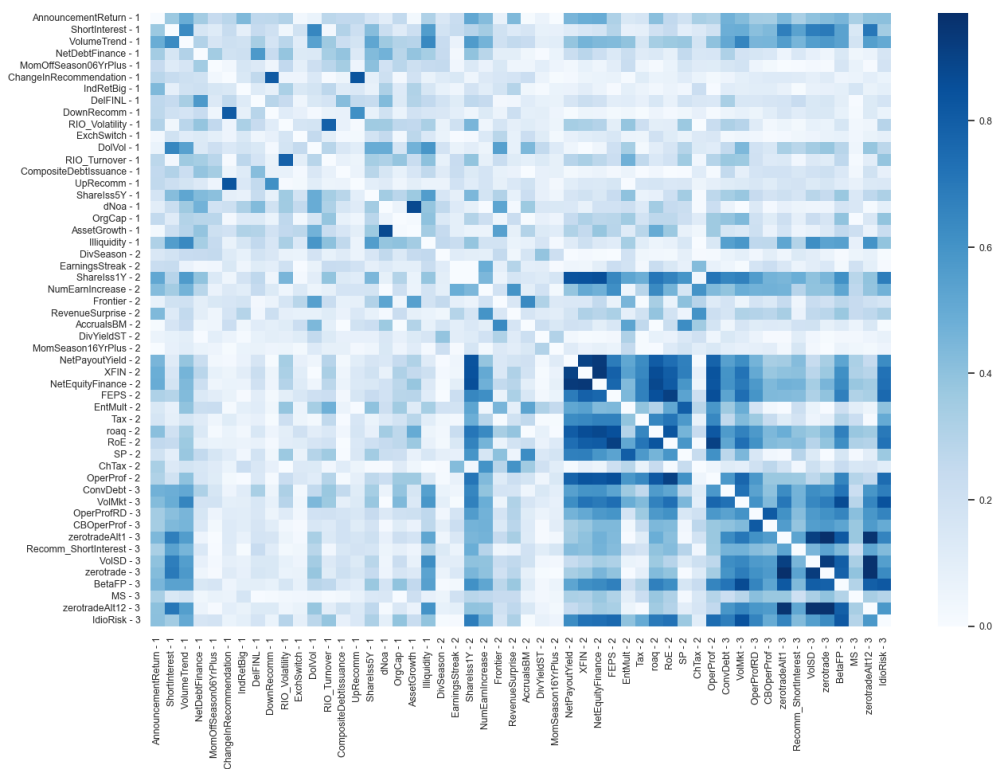


Figure 11: Correlation heat map of factors identified via CAPM in recent sample
The heat map shows the correlation between factors identified via CAPM in recent sample discussed in Section 2.1. The tail number of each factor indicates the cluster number that the factor belongs to. And the diagonal entries are replaced with 0 for better visualization effect.



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Appendix

Appendix A Asymptotic Theory

To derive the asymptotic properties of our estimators, we impose the following regularity conditions. For simplicity, we consider the case where $\gamma = -1$ and the general results can be derived analogously.

Assumption A.1. α -mixing. The vector-valued process $\{\mathbf{f}_t^\top, \mathbf{u}_t^\top\}^\top$ is a stationary α -mixing process with mixing coefficients $\alpha(j) = \sup_{\tau} \sup_{A \in \mathcal{F}_{-\infty}^{\tau}, B \in \mathcal{F}_{\tau+j}^{\infty}} |P(A \cap B) - P(A)P(B)|$, where $\mathcal{F}_{\tau}^s$ is the σ -field generated by $\{\mathbf{f}_t^\top, \mathbf{u}_t^\top\}^\top : \tau \leq t \leq s\}$, and the mixing coefficients satisfy the condition that

$$\sum_{h=1}^{\infty} \alpha(h)^{1-2/\gamma} < \infty,$$

for some $\gamma > 2$.

Assumption A.2. Factor and noise. There exists a positive constant $C < \infty$ such that,

- (a) For all $1 \leq i \leq N$, $1 \leq t \leq T$, $\mathbb{E}[u_{i,t}] = 0$ and $\mathbb{E}u_{i,t}^8 \leq C$.
- (b) $\{\mathbf{u}_t\}$ is a martingale difference sequence. In particular, $\mathbb{E}(\mathbf{u}_t | \mathcal{I}_{t-1}) = 0$, where $\mathcal{I}_t = \{\mathbf{u}_t^\top, \mathbf{u}_{t-1}^\top, \dots, \mathbf{f}_t^\top, \mathbf{f}_{t-1}^\top, \dots\}$.
- (c) Factor and noise are uncorrelated, that is, $\mathbb{E}[e_{i,t} f_{j,s}] = 0$ for any $1 \leq i, j \leq N$, $1 \leq t, s \leq T$.

Assumption A.3. Loading and alpha.

- (a) For each row of $\boldsymbol{\beta}_t^{(j)}$, $\|\boldsymbol{\beta}_{i,t}^{(j)}\| = \mathcal{O}(1)$, and, as $N \rightarrow \infty$, we have $\left\| N^{-1} \boldsymbol{\beta}_t^{(j)\top} \boldsymbol{\beta}_t^{(l)} - \boldsymbol{\Omega}_t^{(j,l)} \right\| \rightarrow 0$ for some $r \times r$ positive definite matrix $\boldsymbol{\Omega}_t^{(j,l)}$ for $j, l = 0, 1, 2$.
- (b) For $1 \leq t \leq T$, $\left\| \frac{1}{\sqrt{N}} \boldsymbol{\beta}_t^\top \mathbb{M}_{1_N} \boldsymbol{\alpha}_t \right\| = \mathcal{O}(1)$.

Assumption A.4. Cross-sectional correlation of noise. There exists some positive constant $C < \infty$ such that,

- (a) $\left\| \mathbb{E}[\mathbf{u}_t \mathbf{u}_t^\top / N] \right\|_1 \leq C$.

(b) For all $1 \leq i \leq N$, $1 \leq t \leq T$, we assume $\sum_{j=1, j \neq i}^N \left| \mathbb{E} \left[u_{t,i} u_{t,j} \right] \right| \leq C$.

(c) For all $1 \leq i, j \leq N$, $1 \leq t \leq T$, we assume

$$\sum_{l=1}^N \sum_{s=1}^T |\text{Cov}(u_{i,t} u_{j,t}, u_{i,s} u_{l,s})| \leq C$$

Assumption A.5. $\beta_{i \cdot}(t/T)$ and $\alpha_{i \cdot}(t/T)$ have continuous derivatives up to the second order.

Moreover, there exists $m > 2$, $1 < a, b < \infty$, $1/a + 1/b = 1$, $c = 0, 1, 2$ such that, for some positive $C < \infty$, $\mathbb{E} \left[\left| \frac{1}{\sqrt{N}} \sum_{i=1}^N u_{i,t} \right|^{mb} \right] = \mathcal{O}(1)$, $\mathbb{E} \left[\left\| \frac{1}{\sqrt{N}} \sum_{i=1}^N \beta_{i,t}^{(c)} u_{i,t} \right\|^{mb} \right] = \mathcal{O}(1)$, and $\mathbb{E} \left[\|f_t\|^{\max(ma, 4)} \right] \leq C$ for any $1 \leq t \leq T$,

Assumption A.6. Kernel. The kernel function $k(\cdot) : [-1, 1] \rightarrow \mathbb{R}^+$ is a symmetric and Lipschitz continuous probability density function such that $\int_{-1}^1 k(u) du = 1$, $\int_{-1}^1 u k(u) du = 0$, and $\int_{-1}^1 u^2 k(u) du < \infty$.

The α -mixing condition in Assumption A.1 allows weak temporal correlations for both the factors and noises. Assumption A.2 mainly imposes moment condition on the error as well as zero correlation between the error and factor. Different from Giglio et al. (2021), both the error and factor are allowed to have serial dependence. Assumption A.3(a) imposes the pervasive condition of the factor loading, following (Stock and Watson, 2002). It ensures that each row of the factor vector f_t has a nontrivial contribution to the variance of r_t . Assumption A.3(b) imposes a mild condition on the relationship between the loading and alpha, which is weaker than Assumption A.4(i) imposed in Fan et al. (2022).

In the high-dimensional factor model, the diverging cross-sectional dimension N also determines the convergence rate of our estimator, which is affected by the cross-sectional dependence. Thus Assumptions A.4 and A.5 are imposed so that the information accumulated over the cross-sectional dimension is useful too. These conditions are obviously satisfied if $\{u_{i,t}\}$ is cross-sectional independent and is independent of f_t , with assumed moment conditions. We include them to allow for weak cross-sectional and temporal dependence and our model is an approximate static factor model similar to Bai and Ng

(2002) and Bai (2003). Assumption A.5 also imposes smoothness conditions on the alpha and beta functions, which have been commonly imposed in the literature (see Robinson (1989), Cai (2007), Su and Wang (2017)). Assumption A.6 is a standard assumption for kernel regressions. This condition is satisfied by widely used second-order kernels, such as the Epanechnikov, uniform, and quartic kernels.

We now state the consistency of $\widehat{\alpha}_t$.

Theorem 1. *Suppose Assumptions A.1-A.6 are satisfied. As k fixed, $h \rightarrow 0$, $Th^3 \rightarrow \infty$, and $T, N \rightarrow \infty$, we have*

$$\frac{1}{N} \|\widehat{\alpha}_t - \alpha_t\|^2 = \mathcal{O}_p\left(\frac{1}{N^2} + \frac{1}{Th} + h^4\right).$$

Theorem 1 establishes the convergence rate of our nonparametric estimator $\widehat{\alpha}_t$, which depends on both T and N . Unlike the latent factor f_t and the factor loading β_t , which are only identified up to a rotation transformation, α_t is identifiable.

Theorem 2. *Suppose Assumptions A.1-A.6 are satisfied. As k fixed, $h \rightarrow 0$, $Th^3 \rightarrow \infty$, $Th^5 \rightarrow 0$, $Th/N \rightarrow 0$, and $T, N \rightarrow \infty$, for each $i = 1, \dots, N$ and $t = 1, \dots, T$, we have*

$$\sqrt{Th}(\widehat{\alpha}_{t,i} - \alpha_{t,i}) \rightarrow^d N(0, \nu_0 \Sigma_{\alpha_{t,i}}),$$

where

$$\Sigma_{\alpha_{t,i}} = \mathbb{E}\left[u_{t,i}^2 \left(1 - \mathbf{v}_t^\top \Sigma_f^{-1} \boldsymbol{\lambda}_t\right)^2\right],$$

$\nu_0 = \int_{-1}^1 k^2(u) du$, and Σ_f is the covariance matrix of f_t .

Theorem 2 shows the asymptotic normality of the nonparametric estimator $\widehat{\alpha}_{t,i}$. As shown in the appendix, the asymptotic bias $\widehat{\alpha}_{t,i}$ is proportional to h^2 in both interior region and boundary regions, thanks to the use of the boundary-modified kernel function in (6). Therefore, when $Th^5 \rightarrow 0$, the asymptotic bias is asymptotically negligible. A candidate bandwidth that satisfies the assumptions is $h = T^{-1/4.5}$ when $T^{3.5/4.5}/N \rightarrow 0$.

Theorems 1 and 2 present the asymptotic properties when the dimension of the latent factor r is assumed to be known. In practice, we can estimate r via the generalized eigenvalue ratio-based estimator defined in (10). The asymptotic validity of $\widehat{r}$ is justified in Theorem 3 below.

Theorem 3. Suppose Assumptions A.1-A.6 are satisfied and $r_{\max}$ is a predetermined constant no smaller than r . Then we have

$$P(\widehat{r} = r) \rightarrow 1$$

as $h \rightarrow 0$, $Th^3 \rightarrow \infty$, $T, N \rightarrow \infty$, where $\widehat{r}$ are defined in (10).

Theorem 3 derives the consistency of the ratio-based estimator $\widehat{r}$. This can be viewed as a generalization of Theorem 1 of Ahn and Horenstein (2013) from constant parameter factor models to time-varying factor models. Other selection criteria, such as BIC (Bai and Ng (2002)) can be adopted too.

Appendix B Proofs

Proof of Theorem 1

Proof. Recall that our estimator $\widehat{\beta}_t$ is given by the matrix of $\sqrt{N}$ times the top k eigenvectors of $\widehat{\Sigma}_t$ in descending order by corresponding eigenvalues. By the definition of eigenvectors and eigenvalues, we have

$$\widehat{\beta}_t = \widehat{\Sigma}_t \widehat{\beta}_t \widehat{V}_t^{-1}.$$

Let

$$\mathbf{H}_t = \frac{1}{NT} \sum_{s=1}^T (\mathbf{v}_s - \bar{\mathbf{v}}_t)(\mathbf{u}_s - \bar{\mathbf{u}}_t)^\top \beta_t^\top \widehat{\beta}_t K_{h,st} \widehat{V}_t^{-1}.$$

Then we have the following decomposition:

$$\begin{aligned}
\widehat{\beta}_t - \beta_t \mathbf{H}_t &= \widehat{\Sigma}_t \widehat{\beta}_t \widehat{\mathbf{V}}_t^{-1} - \beta_t \mathbf{H}_t \\
&= \frac{1}{NT} \sum_{s=1}^T K_{h,st} \left[\beta_t (\mathbf{v}_s - \bar{\mathbf{v}}_t) (\mathbf{u}_s - \bar{\mathbf{u}}_t)^\top + (\mathbf{u}_s - \bar{\mathbf{u}}_t) (\mathbf{v}_s - \bar{\mathbf{v}}_t)^\top \beta_t^\top + (\mathbf{u}_s - \bar{\mathbf{u}}_t) (\mathbf{u}_s - \bar{\mathbf{u}}_t)^\top \right. \\
&\quad + \mathbf{D}_{st} \mathbf{v}_s (\mathbf{v}_s - \bar{\mathbf{v}}_t)^\top \beta_t^\top + \left(\mathbf{D}_{st} \mathbf{v}_s - T^{-1} \sum_{s=1}^T K_{h,st} \mathbf{D}_{st} \mathbf{v}_s \right) \mathbf{v}_s^\top \mathbf{D}_{st}^\top + \beta_t (\mathbf{v}_s - \bar{\mathbf{v}}_t) \mathbf{v}_s^\top \mathbf{D}_{st}^\top \\
&\quad + (\mathbf{u}_s - \bar{\mathbf{u}}_t) \mathbf{v}_s^\top \mathbf{D}_{st}^\top + \mathbf{D}_{st} \mathbf{v}_s (\mathbf{u}_s - \bar{\mathbf{u}}_t)^\top + \alpha_s (\alpha_s - \alpha_t)^\top + \left(\alpha_s (\beta_s \mathbf{v}_s + \beta_s \lambda_s + u_s) \right)^\top \\
&\quad \left. - \alpha_s T^{-1} \sum_{r=1}^T K_{h,rt} (\beta_r \mathbf{v}_r + \beta_r \lambda_r + u_r) \right] \\
&\quad + \left((\beta_s \mathbf{v}_s + \beta_s \lambda_s + u_s) \alpha_s^\top - T^{-1} \sum_{r=1}^T K_{h,rt} (\beta_r \mathbf{v}_r + \beta_r \lambda_r + u_r) \alpha_s^\top \right) \widehat{\beta}_t \widehat{\mathbf{V}}_t^{-1} \\
&\triangleq \sum_{j=1}^{11} \mathbf{I}_j \widehat{\beta}_t \widehat{\mathbf{V}}_t^{-1}, \tag{12}
\end{aligned}$$

where $\mathbf{D}_{st} = \beta_s - \beta_t$ and $\mathbf{v}_t = \mathbf{f}_t - \mathbb{E}[\mathbf{f}_t]$. Then we have $\frac{1}{\sqrt{N}} \|\widehat{\beta}_t - \beta_t \mathbf{H}_t\| = \mathcal{O}_p\left(\frac{1}{\sqrt{Th}} + \frac{1}{N} + h^2\right)$ by Lemma 1 below.

Secondly, we expand $\widehat{\lambda}_t - \mathbf{H}_t^{-1} \lambda_t$:

$$\begin{aligned}
\widehat{\lambda}_t - \mathbf{H}_t^{-1} \lambda_t &= \mathbf{H}_t^{-1} \bar{\mathbf{v}}_t + \frac{1}{N} \widehat{\mathbf{S}}_\beta^{-1} \mathbf{H}_t^\top \beta_t \mathbb{M}_{1_N} \bar{\alpha}_t + \frac{1}{N} \widehat{\mathbf{S}}_\beta^{-1} \widehat{\beta}_t^\top \mathbb{M}_{1_N} \bar{\mathbf{u}}_t \\
&\quad + \frac{1}{N} \widehat{\mathbf{S}}_\beta^{-1} \widehat{\beta}_t^\top \mathbb{M}_{1_N} (\beta_t \mathbf{H}_t - \widehat{\beta}_t) \mathbf{H}_t^{-1} \bar{\mathbf{v}}_t + \frac{1}{N} \widehat{\mathbf{S}}_\beta^{-1} \widehat{\beta}_t^\top \mathbb{M}_{1_N} (\bar{\beta}_t \mathbf{H}_t - \widehat{\beta}_t \bar{\mathbf{K}}_t) \mathbf{H}_t^{-1} \lambda_t \\
&\quad + \frac{1}{N} \widehat{\mathbf{S}}_\beta^{-1} (\widehat{\beta}_t - \beta_t \mathbf{H}_t)^\top \mathbb{M}_{1_N} \bar{\alpha}_t + \frac{1}{N} \widehat{\mathbf{S}}_\beta^{-1} \widehat{\beta}_t^\top \mathbb{M}_{1_N} \left(T^{-1} \sum_{s=1}^T \mathbf{D}_{st} \mathbf{v}_s K_{h,st} \right) \\
&= \mathbf{H}_t^{-1} \bar{\mathbf{v}}_t + \frac{1}{N} \widehat{\mathbf{S}}_\beta^{-1} \mathbf{H}_t^\top \beta_t^\top \mathbb{M}_{1_N} \bar{\alpha}_t + \mathcal{O}_p\left(\frac{1}{N} + \frac{1}{Th} + h^2\right), \tag{13}
\end{aligned}$$

where $\bar{\mathbf{x}}_t = T^{-1} \sum_{s=1}^T \mathbf{x}_s K_{h,st}$, $\mathbf{x} = \mathbf{v}, \mathbf{u}, \mathbf{f}, \alpha$, or β and $\bar{\mathbf{K}}_t = T^{-1} \sum_{s=1}^T K_{h,st}$.

Last, we expand $\widehat{\alpha}_t - \alpha_t$:

$$\begin{aligned}
\widehat{\alpha}_t - \alpha_t &= \bar{\mathbf{u}}_t + \frac{1}{T} \sum_{s=1}^T \beta_s \mathbf{v}_s K_{h,st} + (\bar{\alpha}_t - \alpha_t) - \beta_t \mathbf{H}_t (\widehat{\lambda}_t - \mathbf{H}_t^{-1} \lambda_t) + (\beta_t \mathbf{H}_t - \widehat{\beta}_t) \widehat{\lambda}_t + (\bar{\beta}_t \lambda_t - \beta_t \lambda_t) \\
&= \bar{\mathbf{u}}_t + \left(\frac{1}{T} \sum_{s=1}^T \beta_s \mathbf{v}_s K_{h,st} - \beta_t \bar{\mathbf{v}}_t \right) + (\bar{\alpha}_t - \alpha_t) - \frac{1}{N} \beta_t \mathbf{H}_t \widehat{\mathbf{S}}_\beta^{-1} \mathbf{H}_t^\top \beta_t \mathbb{M}_{1_N} \bar{\alpha}_t \\
&\quad - \frac{1}{N} \beta_t \mathbf{H}_t \widehat{\mathbf{S}}_\beta^{-1} \mathbf{H}_t^\top \beta_t^\top \mathbb{M}_{1_N} \left(T^{-1} \sum_{s=1}^T D_{st} \mathbf{v}_s K_{h,st} \right) - \sum_{j=1}^{11} \mathbf{I}_j \widehat{\beta}_t \widehat{\mathbf{V}}_t^{-1} \widehat{\lambda}_t + \mathcal{O}_p \left(\frac{1}{N} + \frac{1}{Th} + h^2 \right),
\end{aligned} \tag{14}$$

where $\mathbf{I}_j, j = 1, \dots, 8$, is defined in (12). Theorem 1 follows by Lemma 1 below. Note: need the condition $\max_t \left\| \frac{1}{\sqrt{N}} \beta_t^\top \mathbb{M}_{1_N} \alpha_t \right\| = O(1)$ $\square$

Lemma 1. Under Assumption A.1 - A.6, we have for any $1 \leq t \leq T$,

- (a) $\frac{1}{N} \|\mathbf{I}_1 \widehat{\beta}_t\|_F^2 = \mathcal{O}_p \left(\frac{1}{Th} \right);$
- (b) $\frac{1}{N} \|\mathbf{I}_2 \widehat{\beta}_t\|_F^2 = \mathcal{O}_p \left(\frac{1}{Th} \right);$
- (c) $\frac{1}{N} \|\mathbf{I}_3 \widehat{\beta}_t\|_F^2 = \mathcal{O}_p \left(\frac{1}{Th} \right) + \mathcal{O}_p \left(\frac{1}{N^2} \right);$
- (d) $\frac{1}{N} \|\mathbf{I}_j \widehat{\beta}_t\|_F^2 = \mathcal{O}_p \left(h^4 \right), j = 4, 5, 6;$
- (e) $\frac{1}{N} \|\mathbf{I}_j \widehat{\beta}_t\|_F^2 = \mathcal{O}_p \left(h^4 \right), j = 7, 8;$
- (f) $\frac{1}{N} \|\mathbf{I}_9 \widehat{\beta}_t\|_F^2 = \mathcal{O}_p \left(h^4 \right) + \mathcal{O}_p \left(\frac{1}{T^2 h^2} \right);$
- (g) $\frac{1}{N} \|\mathbf{I}_j \widehat{\beta}_t\|_F^2 = \mathcal{O}_p \left(\frac{h^4}{Th} \right), j = 10, 11;$
- (h) $\frac{1}{N} \|\bar{\mathbf{v}}_t\|_F^2 = \mathcal{O}_p \left(\frac{1}{Th} \right);$
- (i) $\frac{1}{N} \|\bar{\mathbf{u}}_t\|_F^2 = \mathcal{O}_p \left(\frac{1}{Th} \right);$
- (j) $\frac{1}{N} \left\| \frac{1}{T} \sum_{s=1}^T \beta_s \mathbf{v}_s K_{h,st} - \beta_t \bar{\mathbf{v}}_t \right\|_F^2 = \mathcal{O}_p \left(\frac{h}{T} \right);$
- (k) $\frac{1}{N} \|\bar{\alpha}_t - \alpha_t\|_F^2 = \mathcal{O}_p \left(h^4 \right);$
- (l) $\frac{1}{N} \left\| \frac{1}{N} \beta_t \mathbf{H}_t \widehat{\mathbf{S}}_\beta^{-1} \mathbf{H}_t^\top \beta_t^\top \mathbb{M}_{1_N} \bar{\alpha}_t \right\|_F^2 = \mathcal{O}_p \left(\frac{1}{N} \right);$

$$(m) \quad \frac{1}{N} \left\| \frac{1}{N} \boldsymbol{\beta}_t \mathbf{H}_t \widehat{\mathbf{S}}_\beta^{-1} \mathbf{H}_t^\top \boldsymbol{\beta}_t^\top \mathbb{M}_{1_N} \left(T^{-1} \sum_{s=1}^T D_{st} \mathbf{v}_s K_{h,st} \right) \right\|_F^2 = \mathcal{O}_p \left(\frac{h}{T} \right);$$

Proof of Theorem 2

Proof.

$$\begin{aligned} \widehat{\alpha}_{t,i} - \alpha_{t,i} &= \bar{\mathbf{u}}_{t,i} + \left(\frac{1}{T} \sum_{s=1}^T \boldsymbol{\beta}_{s,i} \mathbf{v}_s K_{h,st} - \boldsymbol{\beta}_{t,i} \bar{\mathbf{v}}_t \right) + (\bar{\alpha}_{t,i} - \alpha_{t,i}) - \frac{1}{N} \boldsymbol{\beta}_{t,i} \mathbf{H}_t \widehat{\mathbf{S}}_\beta^{-1} \mathbf{H}_t^\top \boldsymbol{\beta}_t^\top \mathbb{M}_{1_N} \bar{\boldsymbol{\alpha}}_t \\ &\quad - \frac{1}{N} \boldsymbol{\beta}_{t,i} \mathbf{H}_t \widehat{\mathbf{S}}_\beta^{-1} \mathbf{H}_t^\top \boldsymbol{\beta}_t^\top \mathbb{M}_{1_N} \left(T^{-1} \sum_{s=1}^T D_{st} \mathbf{v}_s K_{h,st} \right) - \sum_{j=1}^{11} (\mathbf{I}_j)_i \widehat{\boldsymbol{\beta}}_t \widehat{\mathbf{V}}_t^{-1} \widehat{\boldsymbol{\lambda}}_t + \mathcal{O}_p \left(\frac{1}{N} + \frac{1}{Th} + h^2 \right) \\ &\triangleq \sum_{j=1}^{16} \Pi_j, \end{aligned} \tag{15}$$

where $(\mathbf{I}_j)_i$ denotes the i th row of $\mathbf{I}_j$. By Lemma 2 below, we have

$$\begin{aligned} \sqrt{Th}(\widehat{\alpha}_{t,i} - \alpha_{t,i}) &= \frac{1}{\sqrt{Th}} \sum_{s=1}^T K \left(\frac{s-t}{Th} \right) \mathbf{u}_{s,i} \left(1 - \mathbf{v}_s^\top \boldsymbol{\Sigma}_f^{-1} \boldsymbol{\lambda}_s \right) + o_p(1) \\ &\xrightarrow{\mathcal{D}} \mathcal{N}(\mathbf{0}, \nu_0 \boldsymbol{\Sigma}_{\alpha_{t,i}}), \end{aligned}$$

by Lemma 2 and continuous mapping theorem. Note that Here $\nu_0 \boldsymbol{\Sigma}_{\alpha_{t,i}}$ is defined in Theorem 2. $\square$

Lemma 2. Under Assumption A.1 - A.6, we have for any $1 \leq t \leq T$,

- (a) $\Pi_2 = \mathcal{O}_p \left(\frac{h}{T} \right);$
- (b) $\Pi_3 = \mathcal{O}_p \left(h^2 \right);$
- (c) $\Pi_4 = \mathcal{O}_p \left(\frac{1}{\sqrt{N}} \right);$
- (d) $\Pi_5 = \mathcal{O}_p \left(\frac{h}{T} \right);$
- (e) $\Pi_6 = \mathcal{O}_p \left(\frac{1}{Th} \right);$
- (f) $\left\| \Pi_7 - \frac{1}{T} \sum_{s=1}^T \mathbf{K}_{h,st} \mathbf{u}_{s,i} \mathbf{v}_s^\top \boldsymbol{\Sigma}_f^{-1} \boldsymbol{\lambda}_s \right\| = \mathcal{O}_p \left(\frac{1}{Th} \right);$
- (g) $\Pi_8 = \mathcal{O}_p \left(\frac{1}{Th} \right);$

$$(h) \Pi_j = \mathcal{O}_p(h^2), j = 9, 10, 11;$$

$$(i) \Pi_1 = \mathcal{O}_p(h^2) + \mathcal{O}_p\left(\frac{1}{Th}\right);$$

$$(j) \Pi_j = \mathcal{O}_p\left(\frac{h^2}{Th}\right), j = 13, 14;$$

$$(k) \Pi_j = o_p(h^2), j = 15, 16.$$

Proof of Theorem 3

Proof. Let $\Gamma_i = \frac{1}{T} \sum_{t=1}^T \frac{\widehat{\phi}_{i,t}}{\widehat{\phi}_{i+1,t}}$, where $\widehat{\phi}_{i,t}$ is the i th largest eigenvalue of $\widehat{\Sigma}_t$. By Lemma 3 below, we have $\Gamma_i = \mathcal{O}_p(1)$ uniformly for $i = 1, \dots, r-1$. We also have $\Gamma_i = \mathcal{O}_p(1)$, uniformly for $i = r+1, \dots, p-1$ and $\Gamma_r \rightarrow \infty$ by Lemma 4. Then

$$\mathbb{P}(\widehat{r} \leq r) = \mathbb{P}(\Gamma_r > \max(\Gamma_{r+1}, \Gamma_{r+2}, \dots, \Gamma_{rmax})) \rightarrow 1$$

$$\mathbb{P}(\widehat{r} \geq r) = \mathbb{P}(\Gamma_r > \max(\Gamma_1, \Gamma_2, \dots, \Gamma_{r-1})) \rightarrow 1.$$

Therefore, we have $\mathbb{P}(\widehat{r} = r) \rightarrow 1$. □

Lemma 3. For $j = 1, \dots, r-1$,

$$\frac{1}{T} \sum_{t=1}^T \frac{\widehat{\phi}_{j,t}}{\widehat{\phi}_{j+1,t}} = \mathcal{O}_p(1).$$

Lemma 4. , For $j = r, \dots, p-1$,

$$c + o_p(1) \leq \min(\sqrt{qTh}, p) \widehat{\phi}_{j,t} \leq C + o_p(1),$$

where c , C , and $o_p(1)$ are uniform in $r \leq j \leq p-1$ and $1 \leq t \leq T$.